A PRESCRIBED-FACTOR SIEVE AND AN UNCONDITIONAL PRIME-GAP BOUND OF 216

JINGWEI LIU

Abstract. Let $H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n)$. We prove the unconditional bound $H_1 \le 216$ by combining a prescribed-factor equidistribution theorem with a rigorously certified Maynard sieve weight. The analytic theorem treats Type-I convolutions in arithmetic progressions to squarefree moduli $d \le x^{1/2+2\omega}$ possessing an explicit chain of divisors in prescribed scale windows. Its proof replaces global smoothness of the modulus by transport of these divisor witnesses, retains both branches of the genuine terminal scale, and closes the supported q-van der Corput descent and outer dispersion argument without a conjectural input.

The finite sieve component uses a layered support region recording both the number and total mass of coordinates above a fixed threshold. A 208-parameter rational test function for $k = 46$ is verified by exact support arithmetic and 1024-bit outward-rounded Arb integration, giving

$$\frac{46J(F)}{I(F)} > 1.00002665559198621834807922945362904 27.$$

We also certify an admissible 46-tuple of diameter 216. The prescribed-factor theorem, the exact support decomposition, and the Maynard–Polymath transfer then give infinitely many pairs of primes at distance at most 216. The paper distinguishes the cited trace and dispersion theorems from the new support-transport argument; no unproved equidistribution hypothesis is used. All finite data, hashes, interval certificates, and verification programs are supplied as ancillary files.

Contents

Date: September 3, 2026.

2020 *Mathematics Subject Classification.* Primary 11N05, 11N36; Secondary 11L07, 11N35, 65G30.

Key words and phrases. bounded gaps between primes, Maynard–Tao sieve, equidistribution in arithmetic progressions, q-van der Corput method, trace functions, rigorous interval arithmetic.

1. Introduction

Let p_n denote the nth prime and put

$$H_m = \liminf_{n\to\infty}(p_{n+m} - p_n).$$

Zhang's proof that H_1 is finite initiated a rapid sequence of improvements based on distribution estimates for primes in arithmetic progressions and multidimensional sieve weights [11, 8, 2, 3]. The Polymath project obtained the unconditional bound $H_1 \le 246$. Stadlmann subsequently strengthened the smooth-modulus q-van der Corput estimate [9]; a recent preprint proposes combining related equidistribution estimates with a layered sieve support to deduce $H_1 \le 240$ [10].

The purpose of this paper is twofold. First, we prove the exact prescribed-factor Type-I statement needed to enlarge the sieve support. The proof starts from the established dispersion and trace-function estimates of D. H. J. Polymath, replaces every use of global modulus smoothness by an explicit factor or squarefree-support argument, and closes the return from the terminal descendants to the original discrepancy. Second, we construct and rigorously certify a layered $k = 46$ Maynard weight. Together these two components yield the following result.

Theorem 1.1 (Prime-gap bound)**.** *One has*

$$\boxed{H_1 \le 216.}$$

Equivalently, infinitely many pairs of distinct primes differ by at most 216.

The new analytic input is stated next. The terminology concerning coefficient sequences and equidistribution is recalled in section 2.

Definition 1.2 (Prescribed-factor family)**.** For fixed $\omega, \delta, \gamma, \eta > 0$, let $\mathcal{D}_{\mathrm{pf}}(x)$ consist of the integers $d \le x^{1/2+2\omega}$ for which there exist

$$r \mid d, \qquad u \mid d/r, \qquad d_1 \mid r$$

satisfying

$$x^{\gamma-3\eta-\delta} < r < x^{\gamma-3\eta}, \tag{1.1}$$

$$x^{1-\gamma-6\eta-\delta}d^{-1} < u < x^{1-\gamma-6\eta}d^{-1}, \tag{1.2}$$

$$r^2x^{2-\gamma-52\eta-\delta}d^{-4} < d_1 < r^2x^{2-\gamma-52\eta}d^{-4}. \tag{1.3}$$

Theorem 1.3 (Prescribed-factor Type-I estimate)**.** *Let $f = \alpha \star \beta$, where α and β are coefficient sequences at scales M and $N = x^\gamma$, respectively, $MN \asymp x$, $N \le x^{1/2}$, and β has the Siegel–Walfisz property. Suppose*

$$8\omega + 4\delta + 2\gamma < 1, \tag{1.4}$$

$$32\omega + 10\delta - \gamma < 0, \tag{1.5}$$

$$48\omega + 16\delta - 4\gamma < -1. \tag{1.6}$$

There is $\eta_0 = \eta_0(\omega, \delta, \gamma) > 0$ such that, whenever $0 < \eta < \eta_0$, for every $A > 0$ and every integer $a = a(x)$ satisfying $(a, p) = 1$ for every prime $p \le x$,

$$\sum_{\substack{d \in \mathcal{D}_{\mathrm{pf}}(x) \\ d \text{ squarefree}}} \left| \sum_{\substack{x \le n \le 2x \\ n \equiv a \pmod d}} f(n; x) - \frac{1}{\varphi(d)} \sum_{\substack{x \le n \le 2x \\ (n,d)=1}} f(n; x) \right| \ll_{A,\eta,\omega,\delta,\gamma} \frac{x}{(\log x)^A}. \tag{1.7}$$

The estimate is uniform in the permitted residue a, and the constants are uniform when (ω, δ, γ) ranges over a compact subset of the open region above.

Our second input is a finite theorem.

Theorem 1.4 (Layered variational certificate)**.** *For $k = 46$, $R = 521/2000$, $\delta = 1/50$, and $\omega = 57/10000$, there exists a smooth symmetric Maynard test function F supported on the layered region defined in Definition 9.1 such that*

$$\frac{46J(F)}{I(F)} > 1.0000266555919862183480792294536290427. \tag{1.8}$$

The strict inequality is certified by exact rational support arithmetic and outward-rounded 1024-bit interval integration.

1.1. **The correction completed in this paper.** The prescribed-factor statement in Theorem 1.3 appears as Lemma 6 of [10, version 1, p. 12]. The published 2025 theorem [9], however, applies to globally x^{δ}-smooth moduli. Passing to the larger prescribed-factor family requires an additional inheritance argument.

Two concrete issues prevent the extension sketch in [10, version 1, pp. 12, 17–18] from supplying that argument as written. First, the modulus family is stated with a 52η window, restated in the proof with a 100η window, and then used again with 52η. Second, the proof introduces

$$\Delta^* = \min\left\{ \frac{N}{|\Lambda| x^{5\eta}}, \Delta_1 \right\}$$

and sets $\Delta^* = \Delta_1$ using two quantities that are both bounded below by a third. Such bounds do not compare the two quantities and hence do not justify the asserted identity. We discuss these points precisely in section 3. Our repair keeps the 52η family and retains both branches of Δ^* throughout.

No conclusion is drawn from criticism alone. The terminal trace estimate, support transport, supported descent, two-branch exponent ledger, and outer dispersion closure are proved in sections 4 to 8. Their composition proves Theorem 1.3 without using the prescribed-factor claim of [10] as an input.

1.2. **Main novelties.** The paper introduces four ingredients.

(i) A long-scale completion lemma for the terminal two-variable trace sum, uniform for arbitrary squarefree polynomial-size moduli. It uses the second estimate in Polymath Proposition 8.4 and does not require dense divisibility.

(ii) A factor-interface calculus that replaces the two uses of global smoothness by explicit divisors in prescribed windows. Prime supports, units, valuation saturation, q_0-powers, and subpower preprocessing losses are transported separately.
(iii) An explicit one-way inheritance proof. It fixes the order of witness selection, Cauchy duplication, nonnegative enlargement, terminal completion, and outer Cartesian summation. This closes the local and outer transitions without transferring a witness to an unrelated Cauchy copy.
(iv) A layered variational ansatz that records both the count and mass of coordinates above δ. This retains information lost by a purely radial ansatz and produces a certified crossing with $k = 46$ on the corrected support.

1.3. **Computer assistance and logical scope.** The analytic proof is written independently of the verification programs. The programs check finite exponent identities, support incidence relations, and parameter margins; they do not replace the cited trace estimates or the analytic arguments proving Theorem 1.3. Computer assistance is essential for Theorem 1.4. There every input is rational, the support cover is checked by exact linear arithmetic, and all integrals are enclosed by Arb intervals. The candidate and certificates are linked by SHA-256 hashes.

1.4. **Organization.** After the preliminaries, section 3 compares the smooth-modulus parent theorem with the prescribed-factor extension and identifies the two transitions requiring repair. Sections 4–8 prove Theorem 1.3. Section 9 proves the finite certificate, and section 10 combines the analytic and finite components to prove Theorem 1.1. The appendices record the dependency graph and the reproducibility manifest.

2. Preliminaries

Throughout, x tends to infinity. Implied constants may depend on fixed parameters and fixed seminorm bounds, but not on x or on the particular coefficient sequences under consideration. We write $e_q(z) = \exp(2\pi i z/q)$ and use τ for the divisor function.

Definition 2.1 (Coefficient sequences)**.** A coefficient sequence is a function $\alpha : \mathbb{N} \times (1, \infty) \to \mathbb{R}$ satisfying

$$|\alpha(n; x)| \ll \tau(n)^{O(1)}(\log x)^{O(1)}.$$

It is *located at scale* $N = N(x)$ if, for fixed $0 < c < C$, it is supported on $[cN, CN]$. A sequence at scale N has the *Siegel–Walfisz property* if, for all

$A > 0$, all $q, r \geq 1$, and all $(a,q) = 1$,

$$\left| \sum_{\substack{n \equiv a \pmod q \\ (n,r)=1}} \alpha(n;x) - \frac{1}{\varphi(q)} \sum_{(n,qr)=1} \alpha(n;x) \right| \ll_A \tau(qr)^{O(1)} \frac{N}{(\log x)^A}. \tag{2.1}$$

It is *smooth at scale* N if $\alpha(n;x) = \psi_x(n/N)$, where all ψ_x are supported in one fixed compact interval and every fixed derivative is bounded by a fixed power of $\log x$.

For $f = \alpha \star \beta$ put

$$\Delta_f(a;d) = \sum_{\substack{x \leq n \leq 2x \\ n \equiv a \pmod d}} f(n;x) - \frac{1}{\varphi(d)} \sum_{\substack{x \leq n \leq 2x \\ (n,d)=1}} f(n;x). \tag{2.2}$$

Thus equation (1.7) is the assertion

$$\sum_{\substack{d \in \mathcal{D}_{\mathrm{pf}} \\ d \text{ squarefree}}} |\Delta_f(a;d)| \ll_A x(\log x)^{-A}.$$

2.1. **Established analytic inputs.** We use two groups of previously established results.

(i) The sheaf-theoretic complete-sum and Fourier-transform estimates in [2, Theorem 6.17, Corollary 6.24, and Proposition 8.4]. Only the second bound in Proposition 8.4 is used at the terminal stage. Its local rational phase is nondegenerate whenever the two denominator slopes are distinct modulo every prime factor of the squarefree modulus.

(ii) The dispersion identity and initial Type-I reduction in [2, Sections 5 and 8A], in the explicit form reorganized for smooth moduli in [9]. We use the identity before any smoothness-based enlargement and prove the required enlargement for $\mathcal{D}_{\mathrm{pf}}$.

(iii) The algebraic q-van der Corput identities from [9]: Poisson completion, Cauchy–Schwarz, finite Möbius inversion, CRT substitutions, and Taylor expansion of fixed smooth profiles. Every implication needed for the prescribed-factor family is stated and proved below; the smooth-modulus conclusion itself is not used as an input.

The Bombieri–Vinogradov theorem, the prime number theorem, and the standard Maynard–Polymath multidimensional Selberg sieve are used in their customary forms [8, 3].

2.2. **Parameter reserve.** Assume equations (1.4) to (1.6). Set

$$\begin{aligned} \mathfrak{m} = \min\{&1 - 8\omega - 4\delta - 2\gamma, \gamma - 32\omega - 10\delta, 4\gamma - 1 - 48\omega - 16\delta, \\ &\tfrac{1}{2} - 2\omega - \gamma, \gamma - \delta, \gamma - 8\omega - 4\delta, \tfrac{1}{4} - 16\omega - 6\delta\}. \end{aligned} \tag{2.3}$$

Every entry is positive. We fix

$$0 < \eta < \eta_0 := \min\left\{\frac{\delta}{10^{100}}, \frac{\mathfrak{m}}{1600}\right\}, \qquad \varepsilon_{\mathrm{tr}} = \frac{\eta}{1000}. \tag{2.4}$$

The parameter η pays structural losses in the q-van der Corput chain; $\varepsilon_{\mathrm{tr}}$ is reserved for conductor, divisor, and finite-seminorm losses in the terminal trace estimate. Keeping them separate prevents the same power saving from being spent twice.

The first and third inequalities in the parameter region imply

$$12\omega + 6\delta < \gamma < \frac{1}{2} - 2\omega. \tag{2.5}$$

All later auxiliary range conditions follow from equations (1.4) to (1.6) and (2.4).

3. The prescribed-factor extension and the point requiring repair

This section records the precise relation between the published smooth-modulus theorem [9], the S. Type-I extension proposed in [10], and Theorem 1.3. The distinction matters because the larger modulus family is not covered by the published theorem.

3.1. What the published theorem supplies. The 2025 theorem assumes that the relevant modulus divides $P(x^{\delta})$; in particular it is squarefree and every prime factor is at most x^{δ}. Its q-van der Corput argument introduces a divisor at the scale

$$D = \frac{N}{x^{50\eta}H^2}.$$

This theorem and its local reduction are established analytic inputs in the present paper. What must be proved afresh is that the reduction survives when global smoothness is replaced by the two divisor witnesses in the definition of $\mathcal{D}_{\mathrm{pf}}$.

3.2. The window inconsistency. In Lemma 6 on page 12 of version 1 of [10], the S. Type-I statement uses

$$r^2 x^{2-\gamma-52\eta-\delta} d^{-4} < d_1 < r^2 x^{2-\gamma-52\eta} d^{-4}.$$

At the beginning of its proof on page 17 this is restated with 100η, while the factor family immediately following the restatement again uses 52η. These three displayed conditions define different modulus families. The proof therefore does not, as written, establish the stated lemma.

The compatible exponent is 52. Indeed, after the first factorization,

$$H = \frac{x^{\eta} R Q^2}{q_0 M},$$

and the stated divisor window gives

$$\frac{x^{2-\gamma-52\eta}}{Q^4 R^2} \asymp \frac{N}{q_0^2 x^{50\eta} H^2}.$$

Thus $52 = 50 + 2$, with the additional 2η arising from the square of the x^{η} in the definition of H. We retain this window throughout.

3.3. **The terminal minimum.** The more substantive issue occurs on page 18 and concerns

$$\Delta^* = \min\left\{\frac{N}{|\Lambda|x^{5\eta}}, \Delta_1\right\}. \tag{3.1}$$

The extension sketch gives lower bounds of the form

$$\frac{N}{|\Lambda|x^{5\eta}} \gg C, \qquad \Delta_1 \gg C,$$

and then sets $\Delta^* = \Delta_1$. This conclusion does not follow: two quantities being bounded below by the same third quantity gives no ordering between them. In particular, it does not establish $\Delta_1 \leq N/(|\Lambda|x^{5\eta})$.

This is not merely a typographical matter, because reciprocal powers of Δ^* occur in the final estimates. Our repair is to preserve the minimum and use the exact identity

$$\frac{1}{\Delta_1\Delta^*} = \max\left\{\frac{|\Lambda|x^{5\eta}}{N\Delta_1}, \frac{1}{\Delta_1^2}\right\}. \tag{3.2}$$

Both branches are carried through the exponent ledger in section 7.

3.4. **Scholarly claim boundary.** The observations above do not invalidate the published smooth-modulus theorem [9]; they concern only the larger prescribed-factor extension in [10]. Nor do they imply that the desired extension is false. Sections 4–8 give a complete replacement argument: the supported local descent is proved first, and only then consumed by the outer Cartesian and dispersion closure. The attribution used below is therefore: the smooth-modulus q-van der Corput architecture is due to Stadlmann, while the factor-interface transport to $\mathcal{D}_{\mathrm{pf}}$, the genuine-minimum branch analysis, the supported descent, and the corrected outer Cartesian closure are contributions of the present paper.

4. The terminal squarefree trace estimate

Fix $C \geq 1$. Let $\mathcal{P}$ be a class of profiles supported in one fixed compact interval and satisfying

$$\|\psi^{(j)}\|_{L^1} + \|\psi^{(j)}\|_{L^\infty} \leq B_j(\log x)^{A_j}, \qquad 0 \leq j \leq 2, \tag{4.1}$$

for fixed A_j, B_j. Write $\psi_T(n) = \psi((n - x_0)/T)$, allowing an arbitrary real shift x_0. When $T < 1$, this notation denotes a sequence supported on $O_{\mathcal{P}}(1)$ integers and bounded by $(\log x)^{O_{\mathcal{P}}(1)}$.

Lemma 4.1 (Uniform long-scale completion)**.** *Let $q \leq x^C$ be squarefree, and let $K(\cdot; q)$ be a composite trace function whose prime-local sheaves have bounded conductor and no polynomial phase of degree at most one. Uniformly for $1 \leq T \leq x^C$,*

$$\left|\sum_n \psi_T(n)K(n;q)\right| \ll_{\varepsilon_{\mathrm{tr}},\mathcal{P}} x^{\varepsilon_{\mathrm{tr}}}\left(q^{1/2}+\frac{T}{q^{1/2}}\right). \tag{4.2}$$

Proof. Periodize ψ_T modulo q and use the normalized finite Fourier transform

$$\widehat{K}_q(h)=q^{-1/2}\sum_{a \bmod q} K(a;q)e_q(ha).$$

The Fourier-sheaf estimate in [2, Corollary 6.24] gives $|\widehat{K}_q(h)| \ll C_0^{\omega(q)}$ for every h, including zero. Squarefreeness and $q \le x^C$ absorb this factor into $x^{\varepsilon_{\mathrm{tr}}/16}$.

Poisson summation and two integrations by parts give, uniformly for $|y| \le 1/2$,

$$\left|\sum_n \psi_T(n)e(-ny)\right| \ll (\log x)^{O(1)}\frac{T}{(1+T|y|)^2}.$$

Fourier inversion is exact:

$$\sum_n \psi_T(n)K(n;q)=q^{-1/2}\sum_{h \bmod q}\widehat{K}_q(h)\sum_n \psi_T(n)e(-hn/q).$$

The zero frequency is $O(x^{\varepsilon_{\mathrm{tr}}/4}Tq^{-1/2})$, and summation over the nonzero frequencies is $O(x^{\varepsilon_{\mathrm{tr}}/4}q^{1/2})$. The fixed logarithmic factors fit in the remaining reserve. □

At a vanishing denominator we use either zero extension or the projective convention under which a local factor with trivial numerator character is one. The following bound is valid under both conventions.

Lemma 4.2 (Two-variable squarefree trace bound)**.** *Let $m \le x^C$ be squarefree and let $0 < N, \Delta \le x^C$. For arbitrary residue classes $A, B, L, \gamma_1, \gamma_2$ $(\mathrm{mod}\ m)$,*

$$\left|\sum_d\sum_n \psi_\Delta(d)\psi_N(n)e_m\!\left(\frac{AL}{(n+Bd+\gamma_1)(n+(B+L)d+\gamma_2)}\right)\right|$$
$$\ll_{\varepsilon_{\mathrm{tr}},\mathcal{P}} x^{\varepsilon_{\mathrm{tr}}}(AL,m)\left(\frac{N}{\sqrt{m}}+\sqrt{m}\right)\left(\frac{\Delta}{\sqrt{m}}+\sqrt{m}\right). \tag{4.3}$$

Proof. Assume first that $(AL,m)=1$. Complete in n modulo m. At a prime $p \mid m$, CRT gives the local parameter tuple

$$(a_p,b_p,c_p,d_p,e_{h,p})=\left(B,\gamma_1,B+L,\gamma_2,h(AL)^{-1}\right) \pmod p.$$

The two slopes differ by $-L \not\equiv 0 \pmod p$, the numerator character is nontrivial, and the Fourier parameter is unrestricted. The local nondegeneracy hypotheses of [2, Theorem 6.17] are therefore satisfied. Apply Lemma 4.1 to the completed n-sum and then to the d-sum. This is precisely the squarefree, second estimate in [2, Proposition 8.4].

For $g=(AL,m)$, write $m=gm'$; then $(g,m')=1$ and $(AL/g,m')=1$. If D is the product of the two denominators, zero extension gives

$$\mathfrak{e}_m^{(0)}(AL;D)=\mathfrak{e}_{m'}^{(0)}(AL/g;D)\sum_{c\mid g}\mu(c)\mathbf{1}_{c\mid D},$$

whereas the projective convention simply deletes the trivial local factors. For squarefree c, the condition $c\mid D$ occupies at most $2^{\omega(c)}c$ pairs of residue classes modulo c. Substitute $n=cn'+n_0$ and $d=cd'+d_0$ in each pair and apply the coprime estimate with scales $N/c,\Delta/c$ and modulus m'. If a reduced scale is below one, direct counting gives the same four-term majorant. Summing over $c\mid g$ costs $3^{\omega(g)}=x^{o(1)}$ and yields

$$\frac{N\Delta}{m'}+gN+g\Delta+g^2m'\ll g\left(\frac{N}{\sqrt{m}}+\sqrt{m}\right)\left(\frac{\Delta}{\sqrt{m}}+\sqrt{m}\right).$$

This proves the result under either convention. □

Lemma 4.3 (Congruence-restricted trace bound)**.** *Under the hypotheses of Lemma 4.2, let $q\mid m$ and restrict d,n to fixed residue classes modulo q. Then*

$$\left|\sum_{d\equiv d_*\ (q)}\sum_{n\equiv n_*\ (q)}\psi_\Delta(d)\psi_N(n)e_m\left(\frac{AL}{(n+Bd)(n+(B+L)d)}\right)\right| \ll_{\varepsilon_{\mathrm{tr}},\mathcal{P}} \frac{x^{\varepsilon_{\mathrm{tr}}}(AL,m)}{q}\left(\frac{N}{\sqrt{m}}+\sqrt{m}\right)\left(\frac{\Delta}{\sqrt{m}}+\sqrt{m}\right). \tag{4.4}$$

Proof. Write $d=qd'+d_*$, $n=qn'+n_*$, and $m=qm_1$. Since m is squarefree, $(q,m_1)=1$. The q-component is constant; modulo m_1 the phase has the same form, with its numerator multiplied by the unit $\bar{q}^3$. Apply Lemma 4.2 at scales $N/q,\Delta/q$ and modulus m/q. The identity

$$\frac{N/q}{\sqrt{m/q}}+\sqrt{m/q}=q^{-1/2}\left(\frac{N}{\sqrt{m}}+\sqrt{m}\right)$$

in both variables produces the factor $1/q$. Direct counting covers a subunit reduced scale. □

5. Prescribed factors and prime-support transport

After the initial Type-I reduction write a contributing modulus as $d=qr$, where $q\asymp Q$, $r\asymp R$, and put

$$H=\frac{x^\eta RQ^2}{q_0M}. \tag{5.1}$$

The witness $u \mid d/r$, after its common part with q_0 is removed, gives $q_1 = u_1 v_1$ and the ranges

$$\begin{aligned} q_0^{-2} x^{-\delta-5\eta} \frac{Q}{H} \ll U \ll q_0^{-1} x^{-5\eta} \frac{Q}{H}, \\ x^{5\eta} H \ll V \ll q_0 x^{\delta+5\eta} H, \qquad UV \asymp Q/q_0. \end{aligned} \tag{5.2}$$

The second witness and $d \asymp QR$ give

$$d_1 \asymp \frac{x^{2-\gamma-52\eta}}{Q^4 R^2} \asymp \frac{N}{q_0^2 x^{50\eta} H^2}.$$

After the standard subdivision into intervals shorter by $x^{5\eta}$,

$$\frac{N}{q_0^2 x^{\delta+55\eta} H^2} \ll \Delta_1 \ll \frac{N}{q_0^2 x^{55\eta} H^2}. \tag{5.3}$$

5.1. **Reversible small-prime preprocessing.** Put

$$D_0 = \exp((\log x)^{1/3}), \qquad S_x = \exp((\log x)^{2/3}).$$

Lemma 5.1 (Preprocessing)**.** *Let d be squarefree and let $r_0, u_0, d_{1,0}$ witness membership in $\mathcal{D}_{\mathrm{pf}}(x)$. Define*

$$s(d) = \prod_{\substack{p \mid d \\ p \le D_0}} p, \qquad q_{\mathrm{raw}} = d/r_0, \qquad t = (q_{\mathrm{raw}}, s(d)).$$

If $s(d) \le S_x$, then

$$q = q_{\mathrm{raw}}/t, \qquad r = r_0 t$$

gives $d = qr$ and $(q, P(D_0)) = 1$. There exist $u' \mid q$ and $d_{1,0} \mid r$ satisfying, after dyadic localization,

$$S_x^{-1} \frac{x^{1-\gamma-6\eta-\delta}}{QR} \ll u' \ll \frac{x^{1-\gamma-6\eta}}{QR}, \tag{5.4}$$

$$S_x^{-2} \frac{x^{2-\gamma-52\eta-\delta}}{Q^4 R^2} \ll d_{1,0} \ll \frac{x^{2-\gamma-52\eta}}{Q^4 R^2}. \tag{5.5}$$

Proof. Squarefreeness makes the displayed factorization exact. Every small prime formerly in q_{raw} divides t, so the new q is free of primes at most D_0. Set $u' = u_0/(u_0, t)$. Division by at most S_x proves equation (5.4). Since $r_0 = r/t$, insertion into equation (1.3) produces exactly the factor t^{-2} and proves equation (5.5). □

The exceptional set $s(d) > S_x$ is negligible by the standard small-prime-tail estimate in the alternative Type-I reduction. Restoring the preprocessing changes equations (5.2) and (5.3) only by $S_x^{O(1)} = x^{o(1)}$. Every later expression has fixed algebraic depth, so this loss never iterates with x.

Define the independent families

$$\begin{aligned}\mathcal{Q}^\sharp(Q,R) &= \{q \in [Q,2Q] : (q, P(D_0)) = 1, \\ &\qquad\qquad\qquad q \text{ has a divisor in equation (5.4)}\}, \\ \mathcal{R}^\sharp(Q,R) &= \{r \in [R,2R] : r \text{ has a divisor in equation (5.5)}\}.\end{aligned} \tag{5.6}$$

Each nonexceptional modulus lies in one selected pair from $\mathcal{Q}^\sharp \times \mathcal{R}^\sharp$. A witness is selected before any Cartesian enlargement or Cauchy copy is made.

Lemma 5.2 (First-factor extraction)**.** *Let $q^\sharp = q_0 q_1$ be squarefree and let $u \mid q^\sharp$. If*

$$a_0 = (u, q_0), \qquad u_1 = u/a_0, \qquad v_1 = q_1/u_1,$$

then $q_1 = u_1 v_1$ and $u/q_0 \le u_1 \le u$. Consequently the ranges equation (5.2) hold.

Proof. Prime supports in a squarefree number are sets. Removing from u the support shared with q_0 leaves a divisor of q_1; v_1 is its unique complement. The inequalities follow from $a_0 \le q_0$. □

Lemma 5.3 (Second-factor extraction)**.** *Choose one witness $d_1 \mid r$, write $r = d_1 r_1$, and localize $d_1 \asymp \Delta$. Then*

$$S_x^{-2} \frac{N}{q_0^2 x^{\delta+50\eta} H^2} \ll \Delta \ll \frac{N}{q_0^2 x^{50\eta} H^2}.$$

An $O(x^{5\eta})$ smooth cover at scale $\Delta_1 = \Delta/x^{5\eta}$ gives equation (5.3); moreover d_1, r_1 are squarefree and coprime.

Lemma 5.4 (Prime-support transport)**.** *Suppose $q_0 u_1 v_i r$ and $q_0 q_2 r$ are squarefree for $i = 1, 2$, and $(u_1 v_1 v_2, q_0 q_2) = 1$. Let*

$$r = d_1 r_1, \qquad g = (v_1, v_2), \qquad v_i = g v_i^*.$$

Then the supports of

$$d_1, r_1, q_0, u_1, g, v_1^*, v_2^*, q_2$$

are pairwise disjoint. Hence

$$m = r_1 q_0 u_1 [v_1, v_2] q_2 = r_1 q_0 u_1 g v_1^* v_2^* q_2 \tag{5.7}$$

is squarefree, $(d_1, m) = 1$, and $q_0 q_3$ is squarefree for every $q_3 \mid m/q_0$.

Proof. For squarefree integers, divisibility is inclusion of prime supports, gcd is intersection, lcm is union, and coprimality is disjointness. The hypotheses separate all listed supports except the permitted overlap of v_1, v_2; extracting g partitions that last overlap. Splitting $r = d_1 r_1$ proves the remaining assertions. □

Lemma 5.5 (Valuation saturation)**.** *Let*

$$m^* = m^{\lfloor 2\log x\rfloor}, \quad w_2 = (\lambda, m^*) = (\widetilde{\lambda}, m^*), \quad \lambda^* = (\lambda, \widetilde{\lambda}).$$

If $|\lambda|, |\widetilde{\lambda}| < x$*, then, for large* x*,*

$$(\lambda/w_2, m) = (\widetilde{\lambda}/w_2, m) = (\lambda/\lambda^*, m) = (\widetilde{\lambda}/\lambda^*, m) = 1.$$

Proof. For every $p \mid m$, one has $p^{\lfloor 2\log x\rfloor} > x$. Thus the gcd with m^* captures the full p-adic part of each integer. Removing w_2, or the larger common factor λ^*, leaves a unit modulo m. □

Lemma 5.6 (Gcd average without a range hypothesis)**.** *For* $m \geq 1$*,* $K, T > 0$*, and integers* $A \neq 0, B$*,*

$$\sum_{\substack{|k|\leq K \\ (Ak+B,m)\leq T}} (Ak + B, m) \leq \tau(m)\big(2(A, m)K + T\big). \tag{5.8}$$

Proof. Use $(u, m) \leq \sum_{c|(u,m)} c$, interchange the two finite sums, and enlarge equality of the gcd to $c \mid Ak + B$. If soluble, this congruence is one class modulo $c/(A, c)$ and contributes at most $2K(A, c)/c + 1$. Multiplication by c gives at most $2(A, m)K + T$ for every $c \leq T$. There are at most $\tau(m)$ such divisors. □

6. The supported local descent

This section expands the middle of the argument. In particular, none of the implications below is justified by saying that the proof for smooth moduli is unchanged. We retain the support needed at the next stage, state every enlargement in the direction in which it is used, and record its loss.

Fix one nonempty outer localization. Thus Q, R, q_0, H, H^*, U, V satisfy equations (5.1) to (5.3), with $1 \leq H^* \ll H$, and put

$$g = (v_1, v_2), \qquad g_0 = (q_0, \ell), \qquad v_i = g v_i^*, \qquad (v_1^*, v_2^*) = 1. \tag{6.1}$$

All coefficient sequences in this section are bounded by $(\log x)^{O(1)}$. Such factors, as well as a fixed number of dyadic partitions, are written as $x^{o(1)}$. Since η is fixed, each of them is smaller than $x^{\eta/100}$ for sufficiently large x.

6.1. Preparation and the first phase identity.

Choose the witness $d_1 \mid r$ before duplicating any variable, write $r = d_1 r_1$, and localize $d_1 \asymp \Delta$. In the notation of Lemma 5.3,

$$S_x^{-2} \frac{N}{q_0^2 x^{\delta+50\eta} H^2} \ll \Delta \ll \frac{N}{q_0^2 x^{50\eta} H^2}, \qquad \Delta_1 = \Delta x^{-5\eta}. \tag{6.2}$$

For fixed $r_1, u_1, v_1, v_2, q_2, d_0$, let Σ_2 denote

$$\Sigma_2 = \sum_{\substack{d \\ (d,m)=1 \\ d \text{ squarefree}}} \psi_{\Delta_1}(d-d_0) \Bigg| \sum_{\substack{h_1,h_2\sim H^* \\ h_1v_2\neq h_2v_1}} \sideset{}{^\dagger}\sum_{n} C(n;d)\psi_N(n) \\ \times \varphi_1(h_1)\overline{\varphi_2(h_2)}\Psi(n,h_1,h_2;d)\Bigg|, \tag{6.3}$$

where

$$m = r_1q_0u_1[v_1,v_2]q_2, \qquad (n, dr_1q_0u_1v_1v_2) = 1, \qquad (n+\ell dr_1, q_0q_2) = 1. \tag{6.4}$$

The indicator $C(n;d)$ is supported on at most g_0 classes of n modulo q_0 for each class of d modulo q_0.

Lemma 6.1 (Prepared-factor implication)**.** *Uniformly in the fixed outer data, the estimate*

$$\Sigma_2 \ll q_0g_0\Delta N g x^{-10\eta} \tag{6.5}$$

implies

$$\Sigma_1^\sharp \ll g_0RQNUV^2x^{-4\eta}. \tag{6.6}$$

Proof. Grouping the original r-sum according to $r = d_1r_1$, followed by dyadic localization, gives

$$\Sigma_1^\sharp \ll x^{\eta/2}\frac{RUQ}{q_0\Delta}\sup_{\Delta,r_1,u_1,q_2}\Upsilon_0. \tag{6.7}$$

This is a counting step: the number of choices of the frozen variables is $O(x^{\eta/2}RUQ/(q_0\Delta))$ after the usual divisor-weight enlargement. No factor of a prescribed size is chosen here; the factor was already supplied by $d_1(r)$.

Separate the relation $h_1v_2 = h_2v_1$. Since $H^* \ll H \le x^{-5\eta}V$ by equation (5.2), the number of solutions to that relation is $O(x^{o(1)}H^*V)$, and its contribution is

$$\Upsilon_{0,=} \ll x^{o(1)}\Delta H^*VN \ll x^{-5\eta+o(1)}\Delta NV^2. \tag{6.8}$$

For the complementary part, cover $d \asymp \Delta$ by $O(x^{5\eta})$ shifted smooth intervals of length Δ_1. After the outer absolute value has been taken, deleting the original family-membership indicators and the fixed coefficient of modulus at most one can only increase the sum. The remaining squarefree and coprimality conditions are exactly those in equation (6.3). Thus equation (6.5) and

$$\sum_{v_1\asymp V}\sum_{v_2\asymp V}(v_1,v_2) \ll V^2(\log V)^{O(1)} \tag{6.9}$$

give

$$\Upsilon_{0,\neq} \ll q_0 g_0 \Delta N V^2 x^{-5\eta+o(1)}.$$

The right side of equation (6.8) is smaller than this admissible majorant. Insert the result in equation (6.7); the factors q_0 and Δ cancel, and the powers $x^{\eta/2}$ and $x^{-5\eta+o(1)}$ leave at least $x^{-4\eta}$. This proves equation (6.6). □

Lemma 6.2 (CRT phase factorization)**.** *Under the support conditions in Lemma 5.4, there are residue classes $A, B \pmod m$, independent of n, d, h_1, h_2, such that $(A, m) = 1$ and, with $y = h_1 v_2^* - h_2 v_1^*$,*

$$\Psi(n, h_1, h_2; d) = e_d\left(\frac{ay}{nr_1 q_0 u_1 g v_1^* v_2^* q_2}\right) e_m\left(\frac{Ay}{d(n+Bd)}\right). \tag{6.10}$$

One may choose A, B by

$$\begin{aligned} &A \equiv a \pmod{r_1}, && B \equiv 0 \pmod{r_1}, \\ &A \equiv b_1 \pmod{q_0 u_1 g v_1^* v_2^*}, && B \equiv 0 \pmod{q_0 u_1 g v_1^* v_2^*}, \\ &A \equiv b_2 \pmod{q_2}, && B \equiv \ell r_1 \pmod{q_2}. \end{aligned} \tag{6.11}$$

Proof. By Lemma 5.4, the three moduli in equation (6.11) are pairwise coprime and their product is m. The Chinese remainder theorem therefore defines A and B uniquely modulo m. Splitting each additive character of the duplicated phase over the prime-support decomposition and collecting the d-component gives the first factor in equation (6.10); collecting the remaining components gives the second. The residue A is a unit because a, b_1, b_2 are units on their respective support factors. This proves the identity and every inverse appearing in it is legitimate. □

6.2. **The algebraic q-van der Corput step.** Put

$$\mathscr{Y} = \{h_1 v_2^* - h_2 v_1^* : h_1, h_2 \sim H^*\}. \tag{6.12}$$

For a nonzero $y \in \mathscr{Y}$, set

$$w_1 = (y, d), \qquad w_2 = (y, m^*), \qquad m^* = m^{\lfloor 2\log x \rfloor}.$$

After dyadic localization write $w_1 \asymp W_1$ and $y \sim Y$, where

$$1 \ll W_1 \ll \Delta, \qquad 1 \ll |Y| \ll \frac{H^* V}{g}. \tag{6.13}$$

For $w_0 \mid w_1$, define

$$\mathscr{K}_3(n, \widetilde{n}; y, \widetilde{y}, d) := e_m\left(\frac{A\{y(\widetilde{n}+Bd) - \widetilde{y}(n+Bd)\}}{d(n+Bd)(\widetilde{n}+Bd)}\right). \tag{6.14}$$

$$\Sigma_3 := \sum_{\substack{y,\widetilde{y}\in\mathscr{Y}\\ y,\widetilde{y}\sim Y\\ w_1|(y,\widetilde{y})\\ (y,m^*)=(\widetilde{y},m^*)=w_2}} \Bigg| \sum_{\substack{d\\ w_0w_1|d\\ (d/w_1,my\widetilde{y}/w_1^2)=1}} \psi_{\Delta_1}(d-d_0) \times \sum_{\substack{n,\widetilde{n}\\ y\widetilde{n}\equiv\widetilde{y}n\ (d)}}^{\dagger_3} C(n;d)C(\widetilde{n};d)\psi_N(n)\psi_N(\widetilde{n}) \times \mathscr{K}_3(n,\widetilde{n};y,\widetilde{y},d)\Bigg|, \tag{6.15}$$

where $\dagger_3$ retains

$$(n\widetilde{n}, w_1r_1q_0u_1v_1v_2) = 1, \qquad ((n+\ell dr_1)(\widetilde{n}+\ell dr_1), q_0q_2) = 1. \tag{6.16}$$

Lemma 6.3 (Supported q-van der Corput inequality)**.** *For some admissible* W_1, Y, w_0, w_1, w_2,

$$\Sigma_2 \ll x^{3.5\eta+o(1)} g\Delta\, \Sigma_3^{1/2}. \tag{6.17}$$

Consequently

$$\sup \Sigma_3 \ll q_0^2 g_0^2 N^2 x^{-27\eta} \tag{6.18}$$

implies equation (6.5).

Proof. On a fixed d, partition the inner sum by $yn^{-1} \pmod d$. In equation (6.10) the e_d factor then depends only on that residue class and disappears after absolute values. Retain d squarefree until $w_1 = (y,d)$ is fixed. It follows that w_1 is squarefree and $(w_1, m) = 1$.

Only $x^{o(1)}$ values of w_2 occur. Indeed, a nonempty block has $w_2 \le |y| \le x^{C_1}$ for a fixed C_1. If $m = P_1 \cdots P_r$ with $P_1 < \cdots < P_r$, replace each P_j by the j-th prime. This injects the possible divisors of m^* below x^{C_1} into the p_r-smooth integers below that bound. Since $m \le x^{C_*}$ is squarefree, $r = O(\log x/\log\log x)$ and $p_r = O(\log x)$. Rankin's bound with $\sigma = 1/\log p_r$ gives

$$\#\{w_2 : w_2 \mid m^*,\ w_2 \le x^{C_1}\} \le x^{C_1\sigma} \prod_{p\le p_r} (1-p^{-\sigma})^{-1} = x^{o(1)}.$$

Dyadic decomposition in w_1 and y and the sum over residue classes cost at most $x^{2\eta+o(1)}W_1$. Enlarge the squarefree d-sum to the conditions $w_1 \mid d$ and $(d/w_1, w_1) = 1$. The congruence $yn^{-1} \pmod d$ can then be inverted modulo d/w_1. Replacing $(n,d) = 1$ by the weaker $(n, w_1) = 1$ is a nonnegative enlargement.

Cauchy–Schwarz in the residue class and d variables contributes

$$x^{\eta/2}\Delta/W_1.$$

The preceding factor W_1 cancels. The accumulated factor is $x^{2.5\eta+o(1)}\Delta$. The two copied congruences combine to

$$y\widetilde{n} \equiv \widetilde{y}n \pmod{d}.$$

For fixed y, the equation $y = h_1v_2^* - h_2v_1^*$ has at most $O(g)$ solutions in $h_1, h_2 \sim H^*$: all solutions differ by (kv_1^*, kv_2^*). Applying this to y and $\widetilde{y}$ and then taking the square root contributes $gx^{\eta/2+o(1)}$. Finally

$$1_{(d/w_1,w_1)=1} = \sum_{w_0|(d/w_1,w_1)} \mu(w_0)$$

and the divisor count of the squarefree w_1 contributes another $x^{\eta/2+o(1)}$. The resulting inner expression is exactly equation (6.15), proving equation (6.17). Taking the square root of equation (6.18) leaves $q_0g_0Nx^{-13.5\eta}$; multiplication by the prefactor in equation (6.17) gives equation (6.5). $\square$

6.3. **Removal of diagonals and large gcd blocks.** Let

$$t = (w_2, m), \qquad T = \max\{t^{-1}, H^{-1}\}\frac{x^{\delta+100\eta}H^2N}{g\Delta_1}. \tag{6.19}$$

For fixed $y, \widetilde{y}$, let $\Sigma_4(y, \widetilde{y})$ be the inner absolute value in equation (6.15), restricted further by

$$\left(\frac{y(\widetilde{n} + Bd) - \widetilde{y}(n + Bd)}{d}, m\right) \le T. \tag{6.20}$$

Lemma 6.4 (Diagonal and large-gcd removal)**.** *One has*

$$\Sigma_3 \ll \max\left\{\frac{q_0^2x^{\delta+10\eta}H^3}{t}, \frac{H^4}{t}\right\}\sup_{y,\widetilde{y}}\Sigma_4(y, \widetilde{y}) + x^{-47\eta}N^2. \tag{6.21}$$

Proof. By equations (5.2) and (6.13),

$$|Y| \ll \frac{HV}{g} \ll \frac{q_0x^{\delta+5\eta}H^2}{g}. \tag{6.22}$$

If a divisor q of the numerator in equation (6.20) exceeds T, then equations (5.3), (6.19) and (6.22) give

$$\frac{|Y|}{q} \ll q_0^{-1}x^{-150\eta}\frac{\min\{t, H\}}{H^2}. \tag{6.23}$$

We count the discarded tuples in four disjoint cases. If $y = \widetilde{y}$ and $n = \widetilde{n}$, their number is

$$O(\Delta H^2N) \ll q_0^{-2}x^{-50\eta}N^2. \tag{6.24}$$

If $y = \widetilde{y}$ but $n \neq \widetilde{n}$, the congruence with a fixed $q > T$ and equation (6.23) give

$$\max\{H^2/t, H\}N^2\frac{|Y|}{q} \ll q_0^{-1}x^{-150\eta}N^2. \tag{6.25}$$

If $n = \widetilde{n}$ and $y \neq \widetilde{y}$, split according to $(y - \widetilde{y}, dq)$. Divisor expansion bounds the resulting two terms by

$$x^{\delta-144\eta+o(1)}N, \qquad x^{2\delta-39\eta+o(1)}H^2N. \tag{6.26}$$

The inequalities following from equations (1.4) and (1.6), namely $\gamma > \delta$ and $\gamma > 8\omega + 4\delta$, together with $H \ll x^{4\omega+\delta+7\eta}/q_0$, make both terms $O(x^{-48\eta}N^2)$ after reducing η_0.

In the fully off-diagonal case, fix $d, \widetilde{n}, y - \widetilde{y}$. The defining equation leaves $O(q_0 \min\{t, H\})$ possible quotient values and only $x^{o(1)}$ factorizations for each. Hence this contribution is

$$x^{o(1)}q_0 \max\{H^2/t, H\}\Delta N \min\{t, H\} \ll q_0^{-1}x^{-50\eta+o(1)}N^2. \tag{6.27}$$

The number of possible gcd divisors is $x^{o(1)}$, so equations (6.24) to (6.27) sum to the error in equation (6.21).

It remains to count retained phase pairs. The representation through h_1, h_2 gives

$$\#(y, \widetilde{y}) \ll \max\{H^4/t^2, H^2\},$$

whereas the interval bound gives $\#(y, \widetilde{y}) \ll Y^2/t^2$. If $t \leq x^\delta H$, use the first estimate; if $t > x^\delta H$, use the second and equation (6.22). In both cases,

$$\#(y, \widetilde{y}) \ll \max\left\{\frac{q_0^2x^{\delta+10\eta}H^3}{t}, \frac{H^4}{t}\right\}. \tag{6.28}$$

Multiplying the supremum of Σ_4 by this count completes the proof. $\square$

6.4. **Relabelling the retained blocks.** Write

$$z_1 = w_0w_1, \qquad y = w_1\lambda, \qquad \widetilde{y} = w_1\widetilde{\lambda}, \qquad d = z_1e, \qquad \Lambda = Y/w_1. \tag{6.29}$$

An empty or incompatible block is discarded. In every remaining block,

$$z_1 \ll x^{5\eta}\Delta_1, \qquad (z_1, m) = 1, \qquad w_1 \mid z_1, \qquad (\lambda, m^*) = (\widetilde{\lambda}, m^*) = w_2, \tag{6.30}$$

$$H \ll q_0^{-1}x^{4\omega+\delta+7\eta}, \qquad x^{-4\eta-\delta}N \ll R \ll x^{-2\eta}N, \tag{6.31}$$

$$x^{1/2-\eta} \ll QR \ll x^{1/2+2\omega+\eta}, \qquad x^{5\eta}H \ll v_i \ll q_0x^{\delta+5\eta}H, \tag{6.32}$$

$$\frac{RQ^2H}{q_0g\Delta_1} \ll m \ll \frac{x^\delta RQ^2H}{g\Delta_1}, \qquad 1 \ll |\Lambda| \ll \frac{q_0x^{\delta+5\eta}H^2}{w_1g}. \tag{6.33}$$

The apparent asymmetry in the two bounds for m is genuine: the lower endpoint contains the common factor removed from the first prescribed divisor, whereas the upper endpoint uses only its ambient dyadic range.

Replace e again by d, and set

$$\begin{aligned} &l = \ell z_1 r_1, \qquad c_1 = r_1 q_0 u_1 [v_1, v_2], \quad c_2 = q_0 q_2, \\ &A \leftarrow A w_1 / z_1, \quad B \leftarrow B z_1. \end{aligned} \tag{6.34}$$

All quotients in equation (6.34) are units modulo m. The retained sum becomes

$$\Sigma_5 = \Bigg| \sum_{(d, m\lambda\widetilde{\lambda})=1} \psi_{\Delta_1}(z_1 d - d_0) \sum_{n,\widetilde{n}}^{\dagger_5} C(n;d) C(\widetilde{n};d) \psi_N(n) \psi_N(\widetilde{n}) \\ \times e_m\left(\frac{A\{\lambda(\widetilde{n}+Bd) - \widetilde{\lambda}(n+Bd)\}}{d(n+Bd)(\widetilde{n}+Bd)} \right) \Bigg|, \tag{6.35}$$

where $\dagger_5$ means

$$(n\widetilde{n}, w_1 c_1) = 1, \qquad ((n+ld)(\widetilde{n}+ld), c_2) = 1, \tag{6.36}$$

$$\lambda\widetilde{n} \equiv \widetilde{\lambda} n \pmod{z_1 d / w_1}, \qquad \left(\frac{\lambda(\widetilde{n}+Bd) - \widetilde{\lambda}(n+Bd)}{d}, m \right) \le T. \tag{6.37}$$

Equations equations (6.29) and (6.34) are reversible on each nonempty block, so $\Sigma_4 = \Sigma_5$. Combining equations (6.18) and (6.21) shows that it is enough to prove

$$\Sigma_5 \ll \min\left\{ \frac{t}{q_0^2 x^{\delta+10\eta} H^3}, \frac{t}{H^4} \right\} q_0^2 g_0^2 N^2 x^{-27\eta}. \tag{6.38}$$

6.5. **First elimination: the copied variable.** The congruence in equation (6.37) is equivalent to

$$\widetilde{n} = \frac{\widetilde{\lambda} n + k(z_1/w_1)d}{\lambda} \tag{6.39}$$

for an integer k. The supports of the two N-scale profiles and of the Δ_1-scale profile imply

$$|k| \ll K_0 := \frac{w_1 |\Lambda| N}{x^{5\eta} \Delta_1}. \tag{6.40}$$

Put

$$S_k = \frac{z_1}{w_1} k + (\lambda - \widetilde{\lambda}) B. \tag{6.41}$$

For shifted smooth profiles $\psi_N^*, \psi_{\Delta_1}^*$ and residue classes $d_*, n_* \pmod{q_0}$, define

$$\Sigma_6(k) := \left| \sum_{\substack{d\equiv d_*\ (q_0)\\ (d,m\lambda\widetilde{\lambda})=1}} \psi^*_{\Delta_1}(z_1 d - d_0) \sum_{\substack{n\equiv n_*\ (q_0)\\ \lambda|\widetilde{\lambda}n+k(z_1/w_1)d}}^{\dagger_6} \psi^*_N(n) \right.$$
$$\left. \times\, e_m\left(\frac{AS_k}{(n+Bd)\{\lambda^{-1}(\widetilde{\lambda}n+k(z_1/w_1)d)+Bd\}}\right)\right|, \quad (6.42)$$

where $\dagger_6$ retains the four coprimality conditions obtained by substituting equation (6.39) into equation (6.36).

Lemma 6.5 (Elimination of $\widetilde{n}$)**.** *If nonnegative weights ξ_k satisfy*

$$\sum_{\substack{|k|\ll K_0\\ w_2|k\\ (S_k,m)\le T}} \xi_k \ll 1 \quad (6.43)$$

and, uniformly in the profiles and residue classes,

$$\Sigma_6(k) \ll \xi_k \min\left\{\frac{t}{q_0^2 x^{\delta+10\eta}H^3}, \frac{t}{H^4}\right\} q_0 g_0 N^2 x^{-27\eta}, \quad (6.44)$$

then equation (6.38) holds.

Proof. Substitution of equation (6.39) in the phase gives exactly equations (6.41) and (6.42). For the copied smooth weight, write

$$\psi_N\left(\frac{\widetilde{\lambda}n + k(z_1/w_1)d}{\lambda}\right)$$
$$= \sum_{j=0}^{J} \frac{1}{j!}\left(\frac{(k/w_1)(z_1 d - d_0)}{\lambda N}\right)^j \psi^{(j)}\left(\frac{\widetilde{\lambda}n + (k/w_1)d_0}{\lambda N}\right) + O(x^{-100}), \quad (6.45)$$

where $J = \lceil 20/\eta \rceil$. On the support of the sum, the expansion parameter is at most $x^{-5\eta}$ by equation (6.40). Each summand in equation (6.45) is a product of one smooth N-profile in n and one shifted smooth Δ_1-profile in $z_1 d - d_0$. The remainder contributes $O(x^{-90})$ after trivial summation.

For fixed k, the two indicators $C(n; d)$ and $C(\widetilde{n}; d)$ are constant after d, n are fixed modulo q_0. There are at most $q_0 g_0$ contributing pairs of residue classes. Moreover, the divisibility condition in equation (6.42) forces $w_2 \mid k$: both λ and $\widetilde{\lambda}$ are divisible by w_2, whereas $(z_1 d/w_1, w_2) = 1$. Summing equation (6.44) over k and using equation (6.43) supplies the missing factor $q_0 g_0$, and proves equation (6.38). The Taylor remainder is smaller than the target by the fixed parameter reserve. ∎

6.6. **Second elimination: the divisibility by λ.** Let

$$\lambda^* = (|\lambda|, |\widetilde{\lambda}|), \qquad a_\lambda = \lambda/\lambda^*, \qquad b_\lambda = \widetilde{\lambda}/\lambda^*. \tag{6.46}$$

By Lemma 5.5, a_λ and b_λ are units modulo m. We write $\bar{a}_\lambda, \bar{b}_\lambda$ for their inverses modulo m. Likewise, $\overline{\widetilde{\lambda}/w_2}$ denotes the inverse of $\widetilde{\lambda}/w_2$ modulo m.

If $\lambda^* \nmid (z_1/w_1)k$, then equation (6.42) is empty. Otherwise there are residue classes F_k, G_k such that

$$n = a_\lambda n_1 + F_k d, \qquad \widetilde{n} = b_\lambda n_1 + G_k d. \tag{6.47}$$

They satisfy

$$\bar{b}_\lambda(G_k + B) - \bar{a}_\lambda(F_k + B) \equiv \bar{a}_\lambda \, \overline{\widetilde{\lambda}/w_2} \, \frac{S_k}{w_2} \pmod{m}. \tag{6.48}$$

Define the genuine short scale

$$\Delta^* = \min\left\{ \frac{N}{|\Lambda| x^{5\eta}}, \Delta_1 \right\}. \tag{6.49}$$

For shifted profiles at scales Δ^*/z_1 and $|\bar{a}_\lambda|_{\mathrm{sc}} N$ (the latter notation means the real scale $|\lambda^*/\lambda| N$), let

$$\begin{aligned} \Sigma_7(k) := \Bigg| &\sum_{\substack{d \equiv d_* \ (q_0) \\ (d, m\lambda\widetilde{\lambda}) = 1}} \psi_{\Delta^*/z_1}(d) \sum_{n_1}^{\dagger_7} \psi_{|\lambda^*/\lambda| N}(n_1) \\ &\times e_m\!\left(\frac{\bar{a}_\lambda \bar{b}_\lambda A S_k}{(n_1 + \bar{a}_\lambda(F_k + B)d)(n_1 + \bar{b}_\lambda(G_k + B)d)} \right) \Bigg|. \end{aligned} \tag{6.50}$$

Here $\dagger_7$ consists of the congruence $a_\lambda n_1 + F_k d \equiv n_* \pmod{q_0}$ and the four coprimality conditions obtained from equation (6.47).

Lemma 6.6 (Divisibility elimination). *For every retained k,*

$$\Sigma_6(k) \ll \frac{\Delta_1}{\Delta^*} \sup(\Sigma_7(k) + O(x^{-90})), \tag{6.51}$$

where the supremum is over the shifted profiles and the compatible choices of F_k, G_k.

Proof. Dividing the condition $\lambda \mid \widetilde{\lambda} n + (z_1/w_1)kd$ by λ^* gives one linear class for n modulo a_λ because $(a_\lambda, b_\lambda) = 1$. Choosing its representative F_k and substituting gives equation (6.47); the equality defining G_k then gives equation (6.48) by direct subtraction.

Use a smooth partition of unity to split the original d-support into $O(\Delta_1/\Delta^*)$ intervals of length Δ^*/z_1. On one such interval, centered at x_0, the only mixed profile is $\psi_N^*(a_\lambda n_1 + F_k d)$. Its normalized displacement satisfies

$$\frac{|F_k|\,|d-x_0|}{N} \ll \frac{|\lambda|}{\lambda^*}\frac{\Delta^*}{z_1 N} \ll x^{-5\eta}.$$

Taylor expansion to order J separates this profile into products of a shifted n_1-profile and a shifted d-profile; the remainder is $O(x^{-90})$ after summation. The number of partition blocks is the displayed factor in equation (6.51). No replacement of Δ^* by Δ_1 has been made. □

6.7. **Third elimination: all coprimality conditions.** For $q_3 \mid m/q_0$, residue classes $d^*, n^* \pmod{q_0 q_3}$, shifted profiles with $N_2 \le N$ and $\Delta_2 \le \Delta^*$, and units $A_1, A_2 \pmod m$, define

$$\Sigma_8(k) := \Bigg| \sum_{d\equiv d^* \ (q_0q_3)} \ \sum_{n\equiv n^* \ (q_0q_3)} \psi_{\Delta_2/z_1}(d)\psi_{N_2}(n) \times e_m\left(\frac{w_2 A_1 L_1}{(n+B_1 d)(n+(B_1+L_1)d)}\right)\Bigg|, \tag{6.52}$$

where

$$L_1 \equiv A_2 \frac{S_k}{w_2} \pmod m, \qquad (A_1A_2, m) = 1. \tag{6.53}$$

Lemma 6.7 (Coprimality elimination)**.** *For every compatible choice in equation (6.50),*

$$\Sigma_7(k) \ll x^{2\eta+o(1)} q_3 \sup \Sigma_8(k) + O(x^{-89}). \tag{6.54}$$

Proof. We give the full sequence of finite transformations. First apply Mobius inversion to

$$(d, (\lambda/w_2)(\widetilde{\lambda}/w_2)) = 1, \quad (a_\lambda n_1 + F_k d, w_1) = 1, \quad (b_\lambda n_1 + G_k d, w_1) = 1.$$

This introduces

$$f_1 \mid (\lambda/w_2)(\widetilde{\lambda}/w_2), \qquad f_2 \mid w_1, \qquad f_3 \mid w_1, \tag{6.55}$$

and costs $x^{\eta+o(1)}$. Put

$$f_2^* = (f_2, a_\lambda), \qquad f_3^* = (f_3, b_\lambda).$$

Solvability of the two resulting linear congruences first forces a divisor f_{23} of d/f_1. On common prime factors at which their required residues disagree, it forces a further squarefree divisor f_4. The Chinese remainder theorem then gives the exact parametrization

$$\begin{aligned} d &= f^* d_3, & n_1 &= w_3 n_3 + H_k f_4 d_3, \\ f^* &= f_1 f_{23} f_4, & w_3 &= \operatorname{lcm}(f_2/f_2^*, f_3/f_3^*). \end{aligned} \tag{6.56}$$

The supports in equation (6.55) and Lemma 5.5 give $(f^*w_3, m) = 1$ and $w_3 \mid w_1$. Hence every division in equation (6.56) is valid modulo m.

The profile in $w_3n_3 + H_kf_4d_3$ is the last mixed profile. On the d_3-support,

$$\frac{|H_kf_4|\,\Delta^*}{z_1f^*|\lambda^*/\lambda|N} \ll \frac{w_1|\Lambda|\Delta^*}{z_1N} \le x^{-5\eta}.$$

Taylor expansion through order J therefore separates it, with total remainder $O(x^{-89})$.

Apply Mobius inversion a second time to the five remaining conditions. It introduces

$$e_0 \mid m, \qquad e_1, e_2 \mid c_1, \qquad e_3, e_4 \mid c_2 \tag{6.57}$$

and a further $x^{\eta+o(1)}$ loss. Let

$$g_1 = \operatorname{lcm}(q_0, e_0, e_1, e_2, e_3, e_4) = q_0q_3.$$

Because $q_0, c_1, c_2 \mid m$ and m is squarefree, $q_3 \mid m/q_0$. The congruences for d_3, n_3 are either incompatible, in which case the block is empty, or determine at most

$$\frac{g_1}{\operatorname{lcm}(q_0, e_0)}\frac{g_1}{\operatorname{lcm}(q_0, e_1, e_2, e_3, e_4)} \le \frac{g_1}{q_0} = q_3$$

pairs of classes modulo g_1. On each pair, collect the two denominator slopes into B_1 and $B_1 + L_1$. The relation equation (6.48) gives precisely equation (6.53); multiplying by units produces A_1, A_2. The resulting sum is equation (6.52). Multiplying the two Mobius costs and the number of residue pairs proves equation (6.54). □

6.8. **Uniformity of the descendant profiles.**

Lemma 6.8 (Finite-jet closure)**.** *Let $J = \lceil 20/\eta\rceil$ and $J_* = 3J + 2$. Suppose every input profile has fixed compact support and satisfies*

$$\|\psi^{(r)}\|_1 + \|\psi^{(r)}\|_\infty \le B_r(\log x)^{A_r}, \qquad 0 \le r \le J_*. \tag{6.58}$$

Then every terminal profile in equation (6.52) belongs to one common shifted C^2 profile class, independent of all descendant choices. Moreover

$$m \le x^7, \qquad N_2 \le x^7, \qquad \Delta_2/z_1 \le x^7 \tag{6.59}$$

for sufficiently large x.

Proof. Each of the three Taylor expansions differentiates an input profile at most J times. Two terminal derivatives therefore require at most $3J+2 = J_*$ input derivatives. Products, fixed partitions, and affine changes of variable preserve the compact support and yield uniform C^2 seminorms by Leibniz' rule and the chain rule. Every normalized Taylor increment is at most $x^{-5\eta}$, so

$$x^{-5\eta(J+1)}(\log x)^{O_\eta(1)} = O(x^{-100}).$$

There are at most $(J+1)^3$ descendants, a number independent of x.

For the size assertion, each of $r_1, q_0, u_1, v_1, v_2, q_2$ is $O(x)$ and

$$m = r_1 q_0 u_1 [v_1, v_2] q_2 \le x^6.$$

Also $N_2 \le N \le x$ and $\Delta_2/z_1 \le \Delta^* \le \Delta_1 \le \Delta \le x$. The relaxed common exponent seven leaves room for endpoint constants. □

The lemmas in this section form a one-way chain

$$\Sigma_8 \Longrightarrow \Sigma_7 \Longrightarrow \Sigma_6 \Longrightarrow \Sigma_5 \Longrightarrow \Sigma_4 \Longrightarrow \Sigma_3 \Longrightarrow \Sigma_2 \Longrightarrow \Sigma_1^\sharp. \tag{6.60}$$

Every arrow denotes an upper-bound implication, not an asserted equality. Restrictions are deleted only after an absolute value makes enlargement legitimate, and no deleted restriction is later restored.

7. Terminal trace estimate and quantitative closure

We now insert the trace estimate from section 4 into the last sum of the descent. This section also retains both branches of Δ^* and checks all powers of the common factor q_0.

7.1. The raw terminal bound.

For a retained k, put

$$G_k = (S_k, m), \qquad S_k = \frac{z_1}{w_1} k + (\lambda - \widetilde{\lambda}) B. \tag{7.1}$$

Lemma 7.1 (Terminal estimate and cancellation of q_3)**.** *Uniformly over every descendant in equation (6.52),*

$$\Sigma_8(k) \ll \frac{x^{\varepsilon_{\mathrm{tr}}} G_k}{q_0 q_3} \left(\frac{N}{\sqrt{m}} + \sqrt{m} \right) \left(\frac{\Delta^*}{\sqrt{m}} + \sqrt{m} \right). \tag{7.2}$$

Consequently

$$\Sigma_6(k) \ll \frac{x^{3\eta} G_k}{q_0} \frac{\Delta_1}{\Delta^*} \left(\frac{N}{\sqrt{m}} + \sqrt{m} \right) \left(\frac{\Delta^*}{\sqrt{m}} + \sqrt{m} \right). \tag{7.3}$$

Proof. Apply Lemma 4.3 to equation (6.52) with $q = q_0 q_3$. This divisor is squarefree by Lemma 5.4. The numerator is $w_2 A_1 L_1$ and equation (6.53) gives

$$(w_2 A_1 L_1, m) = (A_1 A_2 S_k, m) = G_k.$$

The scale inequalities $N_2 \le N$, $\Delta_2/z_1 \le \Delta^*$ then give equation (7.2).

Insert this estimate into equation (6.54). Its factor q_3 cancels the denominator q_3 exactly; this cancellation occurs before any gcd averaging. Then use equation (6.51). Since $2\eta + \varepsilon_{\mathrm{tr}} + o(1) < 3\eta$ for sufficiently large x, the result is equation (7.3). □

7.2. **The gcd average without $T \geq K$.** The range equation (6.40) and the condition $w_2 \mid k$ lead to $k = w_2 k_1$ and

$$|k_1| \leq K_1 \ll \frac{w_1|\Lambda|N}{x^{5\eta} w_2 \Delta_1}. \tag{7.4}$$

Because $(z_1/w_1, m) = 1$,

$$((z_1/w_1)w_2, m) = (w_2, m) = t. \tag{7.5}$$

The elementary Lemma 5.6 therefore gives

$$\sum_{\substack{|k_1| \leq K_1 \\ G_{w_2 k_1} \leq T}} G_{w_2 k_1} \leq \tau(m)(2tK_1 + T). \tag{7.6}$$

No comparison between K_1 and T is assumed. From equations (6.19) and (6.33),

$$\frac{K_1}{T} \ll S_x^{O(1)} q_0 x^{-100\eta} \frac{\min\{t, H\}}{w_2} \leq S_x^{O(1)} q_0 x^{-100\eta}. \tag{7.7}$$

Define

$$\mathcal{L} = 1 + K_1/T. \tag{7.8}$$

Since $m \leq x^7$, the standard divisor bound gives $\tau(m) \ll_\eta x^\eta$. Dividing equation (7.6) by a sufficiently large constant times $x^\eta t T \mathcal{L}$ produces nonnegative weights ξ_k with total mass at most one. Hence equation (7.3) may be written

$$\Sigma_6(k) \ll \xi_k \mathcal{L} \frac{x^{4\eta} t T}{q_0} \frac{\Delta_1}{\Delta^*} \left(\frac{N}{\sqrt{m}} + \sqrt{m} \right) \left(\frac{\Delta^*}{\sqrt{m}} + \sqrt{m} \right). \tag{7.9}$$

The loss $\mathcal{L}$ appears once. There is no later Cauchy–Schwarz step that squares it.

7.3. **The genuine short scale.** The minimum in equation (6.49) cannot be replaced by Δ_1. Using equation (6.33) gives

$$\frac{N}{|\Lambda| x^{5\eta}} \gg \frac{w_1 g N}{q_0 x^{\delta + 10\eta} H^2}.$$

Together with equation (5.3) and the harmless preprocessing losses this proves, for one absolute C_0,

$$\Delta^* \gg S_x^{-C_0} \frac{N}{q_0^2 x^{\delta + 55\eta} H^2}. \tag{7.10}$$

For the term containing both Δ_1 and Δ^* we use the exact identity

$$\frac{1}{\Delta_1 \Delta^*} = \max\left\{ \frac{|\Lambda| x^{5\eta}}{N \Delta_1}, \frac{1}{\Delta_1^2} \right\}. \tag{7.11}$$

7.4. **The exponent ledger.** Set

$$\mathcal{B} = \min\left\{\frac{1}{q_0^2 x^{\delta+10\eta} H^3}, \frac{1}{H^4}\right\}. \tag{7.12}$$

After substituting T from equation (6.19) into equation (7.9), the bound required in equation (6.44) follows once each of

$$\frac{N\Delta^*}{m}, \qquad N, \qquad m \tag{7.13}$$

is at most

$$\frac{\Delta^*}{\Delta_1}\mathcal{B}\frac{gq_0^2 g_0 N\Delta_1}{x^{\delta+131\eta}H^2}. \tag{7.14}$$

Indeed, expanding the two completion factors gives

$$\left(\frac{N}{\sqrt{m}} + \sqrt{m}\right)\left(\frac{\Delta^*}{\sqrt{m}} + \sqrt{m}\right) = \frac{N\Delta^*}{m} + N + \Delta^* + m,$$

and $\Delta^* \le N$ absorbs the third summand into the second.

After cancelling equation (7.14), define

$$\begin{aligned}
E_1 &= \frac{x^{\delta+131\eta}}{gq_0^2 g_0 m}\max\{q_0^2 x^{\delta+10\eta}H^5, H^6\},\\
E_2 &= \frac{x^{\delta+131\eta}}{gq_0^2 g_0 \Delta^*}\max\{q_0^2 x^{\delta+10\eta}H^5, H^6\},\\
E_3 &= \frac{m x^{\delta+131\eta}}{gq_0^2 g_0 N\Delta^*}\max\{q_0^2 x^{\delta+10\eta}H^5, H^6\}.
\end{aligned} \tag{7.15}$$

Thus it suffices to prove $\mathcal{L}E_j \ll 1$ for $j = 1, 2, 3$.

From the upper bound for Δ_1, the lower bound for m, and $RQ^2 = x^{-\eta}q_0 MH$, one obtains

$$m \gg \frac{q_0^2 x^{54\eta} MH^4}{gN}. \tag{7.16}$$

Using $M \asymp x/N$ and $H \ll q_0^{-1}x^{4\omega+\delta+7\eta}$, the two branches of E_1 are

$$E_1 \ll x^{-1+2\gamma+4\omega+3\delta+94\eta} + x^{-1+2\gamma+8\omega+3\delta+91\eta}. \tag{7.17}$$

The lower bound equation (7.10) gives

$$E_2 \ll S_x^{O(1)}\left(x^{28\omega+10\delta-\gamma+245\eta} + x^{32\omega+10\delta-\gamma+242\eta}\right). \tag{7.18}$$

For E_3, the upper bound for m first gives

$$m \ll \frac{q_0 x^{\delta-\eta} MH^2}{g\Delta_1}. \tag{7.19}$$

The two exact alternatives in equation (7.11) satisfy

$$\frac{|\Lambda|x^{5\eta}}{N\Delta_1} \ll \frac{q_0^3 x^{2\delta+65\eta}H^4}{N^2}, \tag{7.20}$$
$$\frac{1}{\Delta_1^2} \ll \frac{q_0^4 x^{2\delta+110\eta}H^4}{N^2}. \tag{7.21}$$

Substitution in E_3 gives four branches:

$$E_3 \ll S_x^{O(1)}\big(x^{1+44\omega+16\delta-4\gamma+282\eta} + x^{1+48\omega+16\delta-4\gamma+279\eta} + x^{1+44\omega+16\delta-4\gamma+327\eta} + x^{1+48\omega+16\delta-4\gamma+324\eta}\big). \tag{7.22}$$

Before dropping nonpositive factors, the residual powers of q_0 in the eight branches of equations (7.17), (7.18) and (7.22) are

$$-3, -6, -5, -8, -7, -10, -6, -9. \tag{7.23}$$

The second term of $\mathcal{L}$ in equation (7.7) adds one to each power of q_0 and subtracts 100η from its power of x. The powers become

$$-2, -5, -4, -7, -6, -9, -5, -8,$$

and remain nonpositive. Hence $\mathcal{L}$ introduces no new parameter condition.

The η-free exponents in equations (7.17), (7.18) and (7.22) are bounded respectively by

$$-1+8\omega+4\delta+2\gamma, \qquad 32\omega+10\delta-\gamma, \qquad 1+48\omega+16\delta-4\gamma. \tag{7.24}$$

They are strictly negative by equations (1.4) to (1.6). The choice equation (2.4) absorbs every displayed coefficient of η, the fixed $S_x^{O(1)} = x^{o(1)}$ losses, and $\varepsilon_{\mathrm{tr}}$. Therefore $\mathcal{L}E_j \ll 1$ for all three j.

Proposition 7.2 (Uniform duplicated prescribed-factor estimate)**.** *For every admissible outer localization,*

$$\Sigma_1^\sharp \ll g_0 RQNUV^2 x^{-4\eta}, \tag{7.25}$$

uniformly in all outer variables and all compatible descendant choices.

Proof. The exponent ledger proves the pointwise estimate equation (6.44) with the normalized weights from equation (7.9). Apply Lemmas 6.1 and 6.3 to 6.5 in that order. Uniformity of every trace and Taylor constant follows from Lemma 6.8. This yields equation (7.25). □

8. Outer Cartesian and dispersion closure

This section proves that the local estimate Proposition 7.2 applies to the individually witnessed modulus family in Definition 1.2. The order in which witnesses are chosen is part of the argument.

8.1. **Selection and Cartesian enlargement.** Split the modulus range into dyadic blocks $d \asymp D$. The blocks with

$$D \leq x^{1/2}(\log x)^{-B} \tag{8.1}$$

contribute $O_A(x(\log x)^{-A})$ by the Bombieri–Vinogradov part of the external dispersion input (ii), after $B = B(A)$ is chosen. For the remaining blocks, $D \gg x^{1/2-\eta}$ for sufficiently large x.

Delete the moduli for which

$$\prod_{\substack{p \mid d \\ p \leq D_0}} p > S_x. \tag{8.2}$$

The standard small-prime-tail estimate in the same external input bounds their total discrepancy by $O_A(x(\log x)^{-A})$. For every surviving modulus, apply Lemma 5.1. Choose the lexicographically first admissible tuple

$$d \longmapsto (q(d), r(d), u(d), d_1(d)) \tag{8.3}$$

before any variable is duplicated. Localizing q and r gives independent families $\mathcal{Q}^\sharp(Q,R)$ and $\mathcal{R}^\sharp(Q,R)$ as in equation (5.6), with

$$x^{\gamma-\delta-3\eta} \ll R \ll x^{\gamma-2\eta}, \qquad x^{1/2-\eta} \ll QR \ll x^{1/2+2\omega}. \tag{8.4}$$

The strict parameter reserve gives

$$RQ^2 = \frac{(RQ)^2}{R} \ll x^{1+4\omega-\gamma+\delta+3\eta} \ll x. \tag{8.5}$$

Each originally selected pair lies in $\mathcal{Q}^\sharp(Q,R) \times \mathcal{R}^\sharp(Q,R)$. Enlarge to that Cartesian product only after absolute values have made every summand nonnegative. Extra cross-pairs and multiple representations can only enlarge the majorant. Cross-pairs whose product is not squarefree are omitted; every selected original pair remains. No assertion is made that an extra cross-pair inherits either witness.

8.2. **Completion and the common factor.** Let $q^{(1)}, q^{(2)} \in \mathcal{Q}^\sharp(Q,R)$ be the two variables produced by Cauchy–Schwarz and write

$$q_0 = (q^{(1)}, q^{(2)}), \qquad q^{(i)} = q_0 q_i. \tag{8.6}$$

Then $(q_1, q_2) = 1$ and

$$1 \leq q_0 \leq 2Q, \qquad Q/q_0 \leq q_i \leq 2Q/q_0. \tag{8.7}$$

Because every member of $\mathcal{Q}^\sharp$ has no prime factor at most D_0,

$$q_0 = 1 \quad \text{or} \quad q_0 > D_0. \tag{8.8}$$

For the witness $u(q^{(1)})$ chosen in equation (8.3), put

$$u_1 = \frac{u(q^{(1)})}{(u(q^{(1)}), q_0)}, \qquad v_1 = q_1/u_1. \tag{8.9}$$

This is the first-factor extraction in Lemma 5.2; localization gives equation (5.2). If $H < 1$, the completed nonzero-frequency sum is empty. Otherwise localize h, u_1, v_1 to scales H^*, U, V.

For fixed q_0, b_1, b_2 and $0 < |\ell| \ll N/R$, let $\Sigma^\sharp_{H^*,U,V}$ be the completed nonzero-frequency sum before the second Cauchy–Schwarz inequality. Its first Cauchy factor is bounded by

$$\Upsilon_1 \ll \frac{g_0 RQUN}{q_0^2}. \tag{8.10}$$

The second factor is $\Sigma^\sharp_1$, including both copies v_1, v_2, the common variables r, u_1, q_2, and all squarefree and coprimality indicators used in section 6. Consequently equations (7.25) and (8.10) give

$$\begin{aligned}(\Sigma^\sharp_{H^*,U,V})^2 &\ll \frac{g_0 RQUN}{q_0^2}(g_0 RQNUV^2 x^{-4\eta}),\\ \Sigma^\sharp_{H^*,U,V} &\ll \frac{g_0 RQN}{q_0} UV x^{-2\eta} \ll \frac{g_0 RQ^2 N}{q_0^2} x^{-2\eta}.\end{aligned} \tag{8.11}$$

The last inequality uses $UV \asymp Q/q_0$. Summing the fixed number of dyadic and preprocessing localizations changes equation (8.11) to

$$\Sigma^\sharp(Q, R; q_0, b_1, b_2, \ell) \ll \frac{g_0 RQ^2 N}{q_0^2} x^{-2\eta+o(1)} \ll \frac{g_0 RQ^2 N}{q_0^2} x^{-3\eta/2}. \tag{8.12}$$

This is why the local proof keeps a full $x^{-2\eta}$ saving: the outer localizations consume part, but not all, of it.

The completion is taken at

$$H = \frac{x^\eta RQ^2}{q_0 M}.$$

Its modulus is $k = r[q^{(1)}, q^{(2)}] = rq_0q_1q_2 \le 8RQ^2/q_0$. Therefore

$$\frac{H}{kM^{-1+\eta}} \gg (x/M)^\eta \asymp N^\eta \longrightarrow \infty,$$

so the completion range required in the external dispersion theorem is valid. Its discarded-frequency remainder, with a sufficiently large fixed number of profile derivatives, is

$$O_A\left(\frac{MN^2}{R(\log x)^A}\right). \tag{8.13}$$

8.3. **Return to the discrepancy.** Before the final q_0, ℓ summation, the nonzero-frequency contribution is bounded by

$$\sum_{1\le|\ell|\ll N/R} \sum_{q_0\le 2Q} \frac{q_0 M}{RQ^2}\Sigma^\sharp(Q,R;q_0,b_1,b_2,\ell). \tag{8.14}$$

For every nonzero ℓ,

$$\begin{aligned}\sum_{q_0\le 2Q}\frac{(q_0,\ell)}{q_0} &= \sum_{q_0\le 2Q}\frac{1}{q_0}\sum_{c\mid(q_0,\ell)}\varphi(c)\\ &\le (1+\log(2Q))\sum_{c\mid\ell}\frac{\varphi(c)}{c} \ll \tau(|\ell|)\log(2Q) = x^{o(1)}.\end{aligned} \tag{8.15}$$

Insert equation (8.12) into equation (8.14) and use equation (8.15). The result is

$$\ll MNx^{-3\eta/2}\sum_{1\le|\ell|\ll N/R}\sum_{q_0\le 2Q}\frac{(q_0,\ell)}{q_0} \ll \frac{MN^2}{R}x^{-3\eta/2+o(1)} \ll \frac{MN^2}{R}x^{-\eta}. \tag{8.16}$$

This is stronger than equation (8.13) for every prescribed logarithmic power after enlarging its constant.

The zero frequency with $q_0 = 1$ is the main term in the dispersion identity. For $q_0 > 1$, equation (8.8) gives the standard majorant

$$MN(\log x)^{O(1)} + \frac{MN^2}{RD_0}(\log x)^{O(1)}, \tag{8.17}$$

which is absorbed because $R/N \ll x^{-2\eta}$ and D_0 dominates every fixed power of $\log x$. The diagonal $\ell = 0$ is estimated elementarily using equation (8.5) and $R \ll x^{-2\eta}N$.

The initial dispersion identity, input (ii), now converts equations (8.13), (8.16) and (8.17) back to the discrepancy equation (2.2). There are only $(\log x)^{O(1)}$ outer blocks. Adding the low-modulus and exceptional small-prime contributions proves equation (1.7), and therefore Theorem 1.3.

All estimates above are uniform in the permitted initial residue $a = a(x)$. After the initial congruence is fixed, a enters the local argument only through the unit A in equation (6.11); the external complete-sum bounds and every ensuing absolute-value majorant are uniform in that unit.

The complete order of choices is

$$\begin{aligned} d \mapsto (q(d), r(d), u(d), d_1(d)) &\mapsto (Q, R, \mathcal{Q}^\sharp, \mathcal{R}^\sharp),\\ (Q, R, \mathcal{Q}^\sharp, \mathcal{R}^\sharp) \mapsto (q^{(1)}, q^{(2)}, q_0) &\mapsto (u_1, v_1, H^*, U, V).\end{aligned} \tag{8.18}$$

In particular, neither witness is allowed to depend on a later Cauchy copy.

9. The layered variational certificate

This section is a finite, computer-assisted theorem. The analytic equidistribution theorem proved in the preceding sections is not used in the evaluation of the integrals. It enters only when the certified support is connected to the sieve in section 10.

9.1. **The layered support.** Fix

$$k = 46, \qquad R = \frac{521}{2000}, \qquad \delta = \frac{1}{50}, \qquad \omega = \frac{57}{10000}. \tag{9.1}$$

The endpoint used for the marginal integral is

$$A_1 = \frac{1}{4} + \omega = \frac{2557}{10000}, \qquad \varepsilon = R - A_1 = \frac{3}{625}, \qquad R_o = A_1 - \varepsilon = \frac{2509}{10000}. \tag{9.2}$$

Let

m	1	2	3	4	5	6	7	8	9	10
B_m	$\frac{157}{1000}$	$\frac{161}{1000}$	$\frac{179}{1000}$	$\frac{179}{1000}$	$\frac{188}{1000}$	$\frac{199}{1000}$	$\frac{203}{1000}$	$\frac{208}{1000}$	$\frac{208}{1000}$	$\frac{211}{1000}$.

For definiteness put $B_m = 219/1000$ for $m \geq 11$.

Definition 9.1 (Layered support)**.** For $\mathbf{t} = (t_1, \dots, t_k) \in [0,1]^k$, set

$$L(\mathbf{t}) = \{i : t_i > \delta\}, \qquad M(\mathbf{t}) = |L(\mathbf{t})|.$$

The layered region is

$$\Omega = \left\{ \mathbf{t} \in [0,1]^k : \sum_{i=1}^{k} t_i < R, \quad \sum_{i \in L(\mathbf{t})} t_i \leq B_{M(\mathbf{t})} \right\}. \tag{9.3}$$

Only the layers $0 \leq M \leq 10$ are nonempty. Indeed, $10\delta = 1/5 \leq B_{10}$, whereas $11\delta = 11/50 > B_{11}$. The point of equation (9.3) is that it retains both the number of coordinates above δ and their total mass. A radial support condition remembers neither datum.

Proposition 9.2 (Exact support routing)**.** *Every pair of coordinate configurations occurring in the denominator and the truncated marginal functionals on Ω satisfies one of the algebraic support conditions associated with Type I, Type IIa, Type IIb, Type III, Bombieri–Vinogradov, or the prescribed-factor condition of Theorem 1.3. In every prescribed-factor cell, the three strict parameter inequalities in Theorem 1.3 hold with positive rational slack.*

Proof. For fixed values of the two large-coordinate counts, all relevant conditions are rational linear inequalities in the ordered coordinate masses and the auxiliary partition points. Of the $11 \cdot 12/2 = 66$ unordered count pairs, the pair $(0,0)$ contains no large coordinate and is routed directly to the Bombieri–Vinogradov cell. We enumerate admissible factor partitions for each of the remaining 65 pairs and ask Z3 to test the negation of their union in exact rational quantifier-free linear arithmetic. Every returned result is unsatisfiable. The capacities, all covering partitions, and every strict rational margin are recorded in the ancillary support certificate. Thus this is a finite exhaustive check, rather than a sampled-grid calculation. The precise verification interface and its trusted-kernel boundary are described in section B. ☐

9.2. **The rational test function.** Put $x_i = t_i/R$ and define

$$U = \sum_i x_i, \qquad U_b = \sum_{i\in L(\mathbf{t})} x_i, \qquad D = \sum_i x_i^2.$$

For $1 \le m \le 10$ let

$$Y_m = \frac{U_b - m\delta/R}{B_m/R - m\delta/R} \in [0,1]. \tag{9.4}$$

Let $L_j(y) = P_j(2y-1)$ be the shifted Legendre polynomial. The frozen candidate has the form

$$F(\mathbf{t}) = \mathbf{1}_\Omega(\mathbf{t}) \prod_{i=1}^{46} q(x_i) \left(A_{M(\mathbf{t})}(U, Y_{M(\mathbf{t})}) + D\, C_{M(\mathbf{t})}(U) \right), \tag{9.5}$$

where the Y-dependent terms are omitted in layer $M = 0$, and

$$A_m(U,Y) = a_{m,0}(U) + \sum_{\ell=1}^{4} a_{m,\ell}(U) L_\ell(Y). \tag{9.6}$$

Here q is the fixed positive degree-32 one-coordinate polynomial whose 33 shifted-Chebyshev coefficients, with denominator 2^{60}, are frozen in the ancillary profile source identified by hash in section B. All the $a_{m,\ell}$ and C_m are expanded in shifted Legendre polynomials in U. Their degree schedules, indexed by $m = 0, \ldots, 10$, are

$$\begin{aligned} \deg a_{m,0} &= (4,4,4,4,4,4,4,3,2,1,1), \\ \deg a_{m,\ell}\ (\ell \ge 1) &= (-1,4,4,4,4,3,3,2,1,-1,-1), \\ \deg C_m &= (3,3,3,3,3,2,2,1,1,-1,-1), \end{aligned} \tag{9.7}$$

where -1 means that the mode is absent. This gives 208 coefficients. Every coefficient is an integer divided by 2^{50}. The coefficient ordering and the complete numerator vector are part of the candidate whose canonical SHA-256 digest is

$$\begin{gathered} \texttt{fbc1d61a9bb8fd7576219001bee8b3c} \\ \texttt{92728dbfa633d6e710583d8aabd8f98c9}. \end{gathered} \tag{9.8}$$

The digest binds the 208 layer coefficients to $k = 46$, the degree schedules, R_o, and the support digest. The polynomial q is bound separately by the profile-source hash in section B. The floating-point searches that found these frozen inputs are not used in their certification.

9.3. **The interval calculation.** For a square-integrable function supported on Ω, define

$$I(F) = \int_{[0,\infty)^k} F(\mathbf{t})^2 \, \mathrm{d}\mathbf{t}, \tag{9.9}$$

$$J_i(F) = \int_{\substack{\mathbf{t}_{\widehat{i}} \in [0,\infty)^{k-1} \\ \sum_{j\neq i} t_j \leq R_o}} \left(\int_0^\infty F(\mathbf{t}) \, \mathrm{d}t_i \right)^2 \mathrm{d}\mathbf{t}_{\widehat{i}}. \tag{9.10}$$

By symmetry $J_i(F)$ is independent of i; denote it by $J(F)$. After the change of variables $t_i = Rx_i$, write

$$I(F) = R^{46}\overline{I}, \qquad J(F) = R^{47}\overline{J}.$$

Notice that equation (9.10) uses the conservative endpoint R_o. No unverified contribution from its complement is added.

Theorem 9.3 (Certified variational inequality)**.** *The rational function equation (9.5) satisfies*

$$\overline{I} > 9.0507319290958436555533517458740189572 \cdot 10^{-110}, \tag{9.11}$$

$$\overline{J} > 7.5531779869092232940446089973346969994 \cdot 10^{-111}, \tag{9.12}$$

$$46R\overline{J} - \overline{I} > 2.4125261747861770030321563214845716954 \cdot 10^{-114}. \tag{9.13}$$

In particular,

$$\frac{46J(F)}{I(F)} > 1.0000266555919862183480792294536290427 > 1. \tag{9.14}$$

Proof. On each layer the integrands are polynomials over a rational polytope. The exact preprocessing uses rational polynomial arithmetic; the final integrals are evaluated at 1024-bit precision with directed outward rounding in Arb [7]. The I certificate separately encloses all eleven layer contributions. The J certificate separately encloses the small–small, twice-small–large, and large–large channels, and it imports the certified interval for I by its file hash. It then evaluates the signed expression $46R\overline{J} - \overline{I}$ directly, rather than deducing its sign by dividing two rounded decimal approximations. The resulting interval has the positive lower endpoint displayed in equation (9.13). Candidate and support hashes agree in all three artifacts. Full midpoint-radius enclosures and reproduction commands are given in section B. □

9.4. **Removing the polyhedral discontinuity.** The indicator in equation (9.5) is convenient for exact integration, whereas the sieve is usually stated for smooth test functions. The strict margin makes the passage harmless.

Lemma 9.4 (Smooth approximation)**.** *For every $\epsilon > 0$ there is a smooth symmetric function F_ϵ compactly supported in the interior of Ω such that*

$$|I(F_\epsilon) - I(F)| < \epsilon, \qquad |J_i(F_\epsilon) - J_i(F)| < \epsilon \quad (1 \leq i \leq 46).$$

Consequently F_ϵ may be chosen so that $46J(F_\epsilon) > I(F_\epsilon)$.

Proof. The region Ω is a finite union of bounded rational polyhedral cells and its boundary has Lebesgue measure zero. Thus smooth functions compactly supported in the cell interiors are dense in $L^2(\Omega)$. Averaging over the symmetric group preserves the support and gives symmetric approximants. The map

$$T_i G(\mathbf{t}_{\widehat{i}}) = \int_0^\infty G(\mathbf{t})\ \mathrm{d}t_i$$

has operator norm at most $R^{1/2}$ from $L^2(\Omega)$ to $L^2(\mathrm{d}\mathbf{t}_{\widehat{i}})$, by Cauchy–Schwarz on each fibre. Restriction to $\sum_{j\neq i} t_j \leq R_o$ cannot increase the norm. Hence $I(G) = \|G\|_2^2$ and $J_i(G) = \|\mathbf{1}_{\{\sum_{j\neq i} t_j \leq R_o\}} T_i G\|_2^2$ are continuous under L^2 approximation. The positive margin in equation (9.13) therefore persists. □

Together, Proposition 9.2, Theorem 9.3, and Lemma 9.4 prove Theorem 1.4.

10. The sieve interface and the bound $H_1 \leq 216$

For completeness we isolate the precise special case of the Maynard–Polymath sieve used here. This prevents the final implication from depending on the corresponding proposition in [10]. Put

$$W = W(x) = \prod_{p \leq \log\log\log x} p, \qquad B_x = \frac{\varphi(W)}{W}\log x.$$

Also set $\theta(n) = \log n$ for $n \in \mathbb{P}$ and $\theta(n) = 0$ otherwise, and write

$$\Delta_\theta(x; a, Q) = \sum_{\substack{x+O(1)\leq m \leq 2x+O(1) \\ m \equiv a\ (Q)}} \theta(m) - \frac{1}{\varphi(Q)} \sum_{\substack{x+O(1)\leq m\leq 2x+O(1) \\ (m,Q)=1}} \theta(m).$$

If $S \subset [0,\infty)^k$ is compact and downward closed, let $\mathcal{Q}_i(S, \epsilon_0; x)$ be the set of squarefree integers

$$q = \prod_{j\neq i} [d_j, e_j]$$

for which the factors $[d_j, e_j]$ are pairwise coprime and coprime to W, and both vectors $(\log_x d_j)_j$ and $(\log_x e_j)_j$ belong to $(1-\epsilon_0)S$. We say that S has the *common-residue prime-distribution property* if, for every i, $A > 0$, fixed $C > 0$, and integer $a = a(x)$ satisfying $(a, p) = 1$ for every prime $p \leq x$,

$$\sum_{q \in \mathcal{Q}_i(S,\epsilon_0;x)} \tau(q)^C\, |\Delta_\theta(x; a, Wq)| \ll_{A,C,\epsilon_0} \frac{x}{(\log x)^A}. \tag{10.1}$$

The bounded endpoint shifts are uniform. The harmless divisor weight is included so that the hypothesis matches the divisor expansion literally; as usual, it can be recovered from an arbitrarily strong unweighted estimate by divisor-moment truncation.

Proposition 10.1 (Sieve transfer)**.** *Let $\mathcal{H} = \{h_1, \dots, h_k\}$ be admissible, let S be compact and downward closed, and let $G \in C_c^\infty(\operatorname{int} S)$ be symmetric. Assume*

$$\sup_{\mathbf{t},\mathbf{u} \in S} \sum_{j=1}^{k} (t_j + u_j) < 1 \tag{10.2}$$

and suppose that equation (10.1) holds for S. If

$$\sum_{i=1}^{k} J_i(G) > I(G), \tag{10.3}$$

then infinitely many translates $n + \mathcal{H}$ contain at least two primes.

Proof. Choose a residue $b \bmod W$ for which $(b + h_i, W) = 1$ for every i. Use the tensor-antiderivative approximation from [3, Sections 4–5]: after a fixed shrinkage of the support, write G to arbitrary C^1 accuracy as a finite sum of tensor products $\sum_\ell c_\ell \prod_i g_{\ell,i}$ and put $f_{\ell,i}(t) = \int_t^\infty g_{\ell,i}(u)\,du$. With

$$\lambda_f(m) = \sum_{d \mid m} \mu(d) f(\log_x d), \qquad \nu(n) = \left(\sum_\ell c_\ell \prod_{i=1}^{k} \lambda_{f_{\ell,i}}(n + h_i) \right)^2,$$

Theorems 3.4 and 3.5 of [3, 4] give, respectively,

$$\sum_{\substack{x \le n \le 2x \\ n \equiv b \ (W)}} \nu(n) = (I(G) + o(1)) C_x, \tag{10.4}$$

$$\sum_{\substack{x \le n \le 2x \\ n \equiv b \ (W)}} \nu(n)\theta(n + h_i) = (J_i(G) + o(1))\, C_x \log x \qquad (1 \le i \le k), \tag{10.5}$$

where

$$C_x = B_x^{-k} \frac{x}{W} > 0.$$

Condition equation (10.2) is exactly the trivial-support hypothesis in the non-prime asymptotic. For equation (10.5), expansion of the two divisor sums produces the modulus $W \prod_{j \ne i} [d_j, e_j]$. After removing finitely many primes dividing $\prod_{r<s} (h_r - h_s)$ into W, the displayed least common multiples are pairwise coprime. The resulting residue class depends on the divisor tuple, so one must not replace it by a pointwise maximum without proof.

We use the standard rough-residue averaging device. Put $P_x = \prod_{p \le x} p$ and let $\mathcal{A}_i$ be the residue classes modulo P_x satisfying

$$a \equiv b + h_i \pmod{W}, \qquad a \equiv h_i - h_j \pmod{p} \quad \text{for one } j \ne i$$

at each prime $W < p \le x$. Every $a \in \mathcal{A}_i$ is coprime to P_x. For a fixed divisor tuple, the generated class modulo $Wq = W \prod_{j \ne i} [d_j, e_j]$ is represented by

$$\frac{|\mathcal{A}_i|}{(k-1)^{\omega(q)}}$$

members of $\mathcal{A}_i$. Since $(k-1)^{\omega(q)} \le \tau(q)^{C_k}$ for a fixed C_k, its discrepancy is bounded by

$$\frac{\tau(q)^{C_k}}{|\mathcal{A}_i|} \sum_{a\in\mathcal{A}_i} |\Delta_\theta(x; a, Wq)|.$$

After summing the divisor tuples, their multiplicity is another fixed power of $\tau(q)$. Averaging equation (10.1) over $a \in \mathcal{A}_i$ therefore bounds the full error by $O_A(x(\log x)^{-A})$. This is the residue-class transfer used in [10, Section 2.4]. The main term is the one-coordinate deletion integral $J_i(G)$.

By equations (10.3) to (10.5), for all sufficiently large x one has

$$\sum_{\substack{x\le n\le 2x\\ n\equiv b\ (W)}} \nu(n)\left(\sum_{i=1}^{k}\theta(n+h_i) - \log(3x)\right) > 0.$$

Since $\nu(n) \ge 0$ and $n + h_i < 3x$ for large x, some n in the progression has at least two prime translates. The same argument on every sufficiently large dyadic interval gives infinitely many such translates. □

Proposition 9.2 gives the exact support routing. To make the analytic dependencies auditable, table 1 records the estimate used for each named cell. Lemma numbers in the non-BV rows refer to version 1 of [10]; those lemmas are themselves relaxed-modulus variants of the published results shown in the right column. The support certificate verifies the hypotheses

cell	direct input	antecedent or replacement
BV	Bombieri–Vinogradov	[3]
Type I	[10, Lemma 5]	[1, Lemma 5]
Type IIa	[10, Lemma 3]	[2, Theorem 2.8(iv)]
Type IIb	[10, Lemma 4]	[2, Theorem 2.8(ii)]
Type IIc	Theorem 1.3	replaces [10, Lemma 6]
Type III	[10, Lemma 7]	[2, Theorem 2.8(v)]

Table 1. Cell-by-cell analytic inputs for the exact support router.

of these six rows for every pair of cells. The non-Type-IIc rows are the established relaxed-modulus forms cited in the table, while the Type-IIc row is Theorem 1.3, proved in sections 4 to 8. The standard finite convolution decomposition used in [10, Sections 3–4], followed cell by cell through the exact support router, therefore gives equation (10.1). The harmless factor $W = x^{o(1)}$ is absorbed by the small-prime preprocessing. Finally, the inequality $2R < 1$ gives equation (10.2). Hence the smooth approximant furnished by Lemma 9.4 has the common-residue prime-distribution property required by Proposition 10.1.

Lemma 10.2 (A diameter-216 admissible tuple)**.** *The set*

$\mathcal{H} = \{0, 4, 6, 16, 18, 28, 30, 34, 40, 48, 54, 58, 60, 64, 70, 76, 78, 84, 88, 96,$

$$100, 106, 114, 118, 120, 126, 130, 138, 144, 148, 154, 160, 166, 168,$$
$$174, 180, 184, 186, 190, 196, 198, 204, 208, 210, 214, 216\}$$

is an admissible 46*-tuple of diameter* 216*.*

Proof. The entries are distinct and lie in $[0, 216]$. For every prime $p \leq 43$, direct reduction leaves at least one residue class unoccupied; the complete missing-residue table appears in section B. For $p > 46$, a set of 46 integers cannot occupy all p residue classes. Hence no prime sees every residue class and $\mathcal{H}$ is admissible. □

Proof of Theorem 1.1. By Theorem 9.3 and Lemma 9.4, choose a smooth symmetric admissible test function G satisfying

$$\sum_{i=1}^{46} J_i(G) = 46J(G) > I(G).$$

The support inputs in table 1, including the proved Theorem 1.3, verify the distribution property of Proposition 10.1. Apply that proposition to the tuple in Lemma 10.2. Infinitely many intervals of length 216 therefore contain two primes. Replacing those two primes by two consecutive primes lying between them can only decrease their distance, so

$$H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n) \leq 216.$$

□

Remark 10.3 (Logical status)**.** No conjectural equidistribution hypothesis is used in the theorem above. The Type-IIc input is proved as Theorem 1.3; the remaining analytic rows invoke the established results cited in table 1. The $k = 46$ variational inequality and tuple admissibility are finite theorems supported by exact and outward-rounded verification artifacts. Thus the conclusion $H_1 \leq 216$ is unconditional in the usual mathematical sense.

Acknowledgements and computational disclosure

This work developed through an extended human–AI collaboration. The author formulated the prescribed-factor research program, identified the need to retain factor-address information, and obtained the initial diameter-226 construction. The antecedent prescribed-factor framework and its proposed diameter-240 application are due to Stadlmann; the two analytic inheritance issues in that proposal are identified and repaired in this paper. OpenAI Codex proposed the layered address-state variational ansatz that reduced the finite candidate diameter from 226 to 216, helped design and implement the exact-support and interval-arithmetic verification pipeline, assisted in diagnosing and repairing the two source-level inheritance issues treated in section 3, and supported the organization and preparation of the manuscript. The author reviewed the resulting arguments and artifacts and accepts full responsibility for every mathematical claim, citation, and conclusion in the

paper. The finite certificates and source code are supplied so that the computer-assisted component can be checked independently.

Appendix A. Corrections, attribution, and verification

The logical dependency graph of the argument is

$$\begin{gathered}\text{published dispersion and trace-function inputs,}\\ \text{the prescribed-factor and support lemmas of section 5,}\\ \text{the expanded local descent of section 6,}\\ \text{the branch ledger of section 7}\\ \Downarrow\\ \textit{Proposition } 7.2 \quad\Longrightarrow\quad \textit{Theorem } 1.3.\end{gathered} \tag{A.1}$$

The final equidistribution conclusion is never used to prove a local estimate. In particular, the outer Cartesian enlargement in section 8 consumes Proposition 7.2 only after the latter has been proved. Thus equation (A.1) is acyclic.

The argument uses three kinds of material, which should not be conflated.

(1) The Deligne-type complete-sum estimates and the initial dispersion identity are cited established theorems, not reproved here.

(2) The order of the q-van der Corput transformations originates in Stadlmann's argument. The exact prescribed-factor family with its 52η window was recorded in [10]. We retain both attributions even though the middle calculation has been rewritten rather than incorporated by reference.

(3) The corrected prescribed-factor family, reversible small-prime preprocessing, prime-support transport, full q_0 ledger, genuine Δ^* bifurcation, and gcd average without $T \geq K$ are the modified parts proved in this document.

For reproducibility, the following finite checks accompany the manuscript:

check	purpose
`supplement/scripts/check_bounded_gap_s52_support_transport.py`	squarefree support and CRT units
`supplement/scripts/check_bounded_gap_s52_reversible_preprocessing.py`	witness transport under preprocessing
`supplement/scripts/check_bounded_gap_s52_q0_transport.py`	full q_0 loss transport
`supplement/scripts/check_bounded_gap_s52_gcd_repair.py`	finite audit of the gcd-average repair
`supplement/scripts/check_bounded_gap_s52_exponent_ledger.py`	rational exponent and q_0 bookkeeping
`supplement/scripts/check_bounded_gap_s52_outer_closure.py`	outer Cauchy and dispersion monomials

These programs are regression checks only; none replaces an analytic estimate or a proof above.

Relative to the established inputs (i)–(iii), the displayed argument proves Theorem 1.3; no conjectural estimate is used. For clarity, the proof has been organized around the following four verification checkpoints:

(V1) verify that the stated form of the squarefree trace estimate follows from the cited sheaf results under both denominator conventions;
(V2) compare every support deletion in section 6 against the frozen source calculation, including empty subunit ranges;
(V3) independently recompute the eight exponent branches and the $\mathcal{L}$ correction;
(V4) verify the exact hypotheses of the external dispersion reduction for the Cartesian families in section 8.

These checkpoints record where each cited input enters; they are not additional hypotheses of Theorem 1.3, and the finite programs above are not substitutes for the analytic argument. This paper establishes the prescribed-factor Type-I estimate itself and then supplies the separate sieve optimization and exact support check required for its bounded-gap application. Those components are combined in section 10 to prove Theorem 1.1.

Appendix B. Finite certificates and reproduction

B.1. **Frozen artifacts.** The finite verification uses the five output artifacts and the separately hashed one-coordinate profile source in table 2. The first column is the SHA-256 digest of the file bytes. The canonical candidate digest in equation (9.8) is different: it hashes only the mathematical binding object and is stable under harmless JSON formatting changes.

Frozen file and file-level SHA-256 digest
`bounded_gap_216_address_state_rational_candidate.json` `c99e6bfc63fd530250ab3d439ff688c9e9c4eb9243bf04c37d6b895f15885fb2`
`bounded_gap_216_address_state_support_exact.json` `18c4c52703f3415d6cb59e02259cb832f700a0210fbc5d46dd4952417964ed0c`
`bounded_gap_216_address_state_i_arb.json` `7a95817358ecace2879270aee77926795bcaeeb80ab38f7bf7f545bea0d9f405`
`bounded_gap_216_address_state_j_arb.json` `fa7a7124992aba6014483ed10ec8908b26ad598579933bf6f1dbd29049029556`
`bounded_gap_216_single_assumption_audit.json` `4181126c1c84f366f03ea43894fba53afce2c0aeff4828cbe44ff844104d83b5`
`build_bounded_gap_236_rational_candidate.py` `e3d44a26ec147c5eca0b04491ee044a9ea41d6b20099364bf2e98a340d2b3bcd`

Table 2. Finite-certificate and profile-source manifest.

The ancillary package also contains the machine-readable file `MANIFEST.sha256`, which covers `requirements.txt`, the ancillary README, every supplied Python source, and every frozen JSON output. Its SHA-256 digest is

`ce39da06a82645414027c5a0c701982c`

`eefb5898c171a4f3443806a8cdad9f3a`.

Generated virtual environments, bytecode, and cache directories are deliberately excluded from the package.

The support calculation is performed in exact rational arithmetic. Its current proof kernel trusts the QF-LRA unsatisfiability answers returned by Z3 [5]; the JSON records the complete rational input and the covering partitions, but not a separately replayable solver proof object. The integral calculation uses FLINT rational polynomials [6] followed by Arb balls with directed outward rounding [7]. No floating-point optimizer output enters the proof.

B.2. **Reproduction commands.** From the ancillary package root, create an isolated environment and install the versions recorded in `requirements.txt`:

```
shasum -a 256 --check MANIFEST.sha256
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
```

The frozen finite candidate is then independently replayed and certified by

```
PYTHONPATH=scripts .venv/bin/python \
  scripts/check_bounded_gap_226_layered_support.py \
  --k 46 \
  --candidate-json \
  outputs/bounded_gap_216_address_state_rational_candidate.json
      \
  --output \
  outputs/bounded_gap_216_address_state_support_exact.json

PYTHONPATH=scripts .venv/bin/python \
  scripts/certify_bounded_gap_216_address_state_i_arb.py \
  --precision 1024

PYTHONPATH=scripts .venv/bin/python \
  scripts/certify_bounded_gap_216_address_state_j_arb.py \
  --precision 1024

PYTHONPATH=scripts .venv/bin/python \
  scripts/audit_bounded_gap_216_single_assumption.py
```

The filename of the generic support checker retains the earlier `226` development label; the explicit `-k 46` and candidate arguments bind this run to the present certificate. The floating-point optimization history that discovered the frozen candidate is not part of the ancillary package. The absolute path in its provenance field is descriptive metadata only and is excluded from the canonical mathematical digest. Reproduction here means replaying the exact support and interval certificates from the frozen rational input, not reproducing the heuristic search that found it.

The analytic regression checks are

```
.venv/bin/python scripts/check_bounded_gap_s52_exponent_ledger.
    py
.venv/bin/python scripts/check_bounded_gap_s52_q0_transport.py
.venv/bin/python scripts/check_bounded_gap_s52_gcd_repair.py
.venv/bin/python scripts/
    check_bounded_gap_s52_reversible_preprocessing.py
.venv/bin/python scripts/
    check_bounded_gap_s52_support_transport.py
.venv/bin/python scripts/check_bounded_gap_s52_loss_budget.py
.venv/bin/python scripts/
    check_bounded_gap_s52_finite_jet_closure.py
.venv/bin/python scripts/check_bounded_gap_s52_outer_closure.py
```

These scripts verify finite algebraic identities and artifact bindings. They do not replace the analytic arguments in sections 4 to 8.

B.3. **Admissibility table.** For the tuple in Lemma 10.2, the following residue classes are unoccupied. One missing class for each prime would suffice; all classes reported by the audit are retained here.

p	missing residues	p	missing residues
2	1	3	2
5	2	7	3
11	2	13	7
17	5	19	17
23	21	29	$8, 17$
31	$1, 15$	37	$8, 24, 35$
41	$1, 8, 12, 27, 33, 39$	43	$3, 7, 13, 22, 23, 29$

For every prime $p > 46$, cardinality alone supplies an unoccupied class.

B.4. **Source provenance.** For auditing the extension against the source arguments, the frozen source snapshots used in development had digests

Polymath equidistribution source
`fdffe1dfb7b820d8a45ecc0e07e2f7e`
`17404e6e10b63db110c2d44afe42013ea`

Stadlmann smooth-modulus source
`60c0440f33d9cbf504470716491fb4d4`
`5b45b26d9a960c8e34ff2af500837a30`

Stadlmann 2026 version-1 source
`c0d5d2317c77f4de7eacdef6e1d4b1e`
`b6433e6240b5c09273b3d4eee99e6c3ba`

The first two rows are established inputs. The third is used to identify and compare the proposed relaxed-modulus estimates and their support conditions. The present paper proves the corrected prescribed-factor theorem Theorem 1.3; the source snapshot is not used as authority for that conclusion.

References

1. Roger C. Baker and Alastair J. Irving, *Bounded intervals containing many primes*, Mathematical Proceedings of the Cambridge Philosophical Society **163** (2017), no. 3, 575–586.
2. D. H. J. Polymath, *New equidistribution estimates of Zhang type*, Algebra & Number Theory **8** (2014), no. 9, 2067–2199.
3. ———, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Research in the Mathematical Sciences **1** (2014), Article 12.
4. ———, *Erratum to: Variants of the Selberg sieve, and bounded intervals containing many primes*, Research in the Mathematical Sciences **2** (2015), Article 15.
5. Leonardo de Moura and Nikolaj Bjørner, *Z3: An efficient SMT solver*, Tools and Algorithms for the Construction and Analysis of Systems, Lecture Notes in Computer Science, vol. 4963, Springer, 2008, pp. 337–340.
6. William B. Hart, *FLINT: Fast library for number theory—an introduction*, Mathematical Software—ICMS 2010, Lecture Notes in Computer Science, vol. 6327, Springer, 2010, pp. 88–91.
7. Fredrik Johansson, *Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Transactions on Computers **66** (2017), no. 8, 1281–1292.
8. James Maynard, *Small gaps between primes*, Annals of Mathematics **181** (2015), no. 1, 383–413.
9. Julia Stadlmann, *On primes in arithmetic progressions and bounded gaps between many primes*, Advances in Mathematics **468** (2025), 110190.
10. ———, *Bounded gaps between primes*, 2026, arXiv:2608.31126v1 [math.NT].
11. Yitang Zhang, *Bounded gaps between primes*, Annals of Mathematics **179** (2014), no. 3, 1121–1174.